



To trust or not to trust a human(-like) AI—A scoping review and conjoint analyses on factors influencing anthropomorphism and trust

Muriel Reuter¹ · Britta Marleen Kirchhoff¹ · Thomas Franke² · Thea Radüntz¹ · Corinna Peifer³

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Abstract

AI systems are becoming increasingly complex and human-like, and we interact with them more and more frequently. How does perceived human-likeness affect trust in AI systems? And what makes AI systems appear human in the first place? In a scoping review, we first examined the relationship between anthropomorphism and trust, although the operationalisation of anthropomorphism was very inconsistent. To address this gap, two conjoint analyses were conducted online focusing on four anthropomorphic characteristics identified in the review: name, appearance, voice, and communication style. The studies found that voice and communication style significantly influenced perceptions of human-likeness, while voice had a slightly stronger effect on trustworthiness. Overall, more human-like systems were perceived as more trustworthy across all attributes.

Practical Relevance: The findings highlight the need for a comprehensive, integrated approach to AI design that considers how design elements shape user perceptions and trust. Importantly, the context in which AI is used, particularly in the workplace, must always be considered.

Keywords Human-computer interaction · Anthropomorphism · Trust · AI systems

✉ Muriel Reuter, M.Sc.
reuter.muriel@baua.bund.de

Dr. rer. nat. Britta Marleen Kirchhoff
kirchhoff.britta@baua.bund.de

Prof. Dr. rer. nat. Thomas Franke
thomas.franke@uni-luebeck.de

PD Dr. rer. nat. Thea Radüntz
raduentz.thea@baua.bund.de

Prof. Dr. rer. nat. Corinna Peifer
corinna.peifer@uni-luebeck.de

¹ Federal Institute for Occupational Safety and Health (BAuA),
Friedrich-Henkel-Weg 1–25, 44149 Dortmund, Germany

² Institute for Multimedia and interactive Systems, University
of Lübeck, Ratzeburger Allee 160, 23562 Lübeck, Germany

³ Research Group Work and Health, Department of
Psychology I, University of Lübeck, Maria-Goeppert-Straße
9a, 23562 Lübeck, Germany

Vertrauen oder Nichtvertrauen gegenüber einer (mensenähnlichen) KI – Ein Scoping-Review und Conjoint-Analysen zu Faktoren, die Anthropomorphismus und Vertrauen beeinflussen

Zusammenfassung

KI-Systeme werden zunehmend komplexer und menschlicher und immer häufiger interagieren wir mit ihnen. Wie wirkt sich die wahrgenommene Menschlichkeit auf das Vertrauen in das KI-System aus? Und was lässt KI-Systeme überhaupt menschlich wirken? In einem Scoping-Review untersuchten wir zunächst die Beziehung zwischen Anthropomorphismus und Vertrauen, wobei die Operationalisierung von Anthropomorphismus sehr uneinheitlich war. Um diese Lücke zu schließen, wurden online zwei Conjoint-Analysen durchgeführt, die sich auf vier anthropomorphe Merkmale konzentrierten: Name, Aussehen, Stimme und Kommunikationsstil. Die Studien ergaben, dass Stimme und Kommunikationsstil die Wahrnehmung der Menschenähnlichkeit deutlich beeinflussten, während die Stimme einen etwas stärkeren Einfluss auf die Vertrauenswürdigkeit hatte. Insgesamt wurden menschenähnlichere Systeme über alle Attribute hinweg als vertrauenswürdiger wahrgenommen.

Praktische Relevanz: Die Ergebnisse unterstreichen die Notwendigkeit eines holistischen Ansatzes für die Entwicklung von KI-Systemen, der berücksichtigt, wie Design-Elemente – wie z.B. Anthropomorphismus – die Wahrnehmung und das Vertrauen der Nutzer beeinflussen. Wichtig ist weiterhin, dass der Kontext, in dem KI eingesetzt wird, insbesondere am Arbeitsplatz, immer berücksichtigt werden muss.

Schlüsselwörter Mensch-Computer Interaktion · Anthropomorphismus · Vertrauen · KI-Systeme

1 Introduction

With the rise of generative artificial intelligence (AI) and the emergence of advanced AI agents, the role of artificial intelligence is shifting beyond that of a conventional tool (Berretta et al. 2023), raising significant concerns for safety, health and well-being in digital work environments. In this dynamic interaction, AI assumes a novel role that calls for thoughtful reflection, as it enhances and transforms human agency instead of simply substituting it (Adolph et al. 2025). Against this backdrop, AI-based support systems have gained prominence, offering the potential to enhance human decision-making processes and increase operational efficiency across various domains. Their presence is for example particularly prominent and consequential in control room settings, where operators are responsible for monitoring intricate systems and making high-stakes decisions with potentially wide-ranging implications.

A key factor in the effective and safe adoption, integration as sociotechnical systems, and acceptance of such technologies is user's trust. While trust has a long research history in the field of human-machine interaction, it needs a critical reassessment in the light of the AI evolution. Two definitions of trust dominate research in this field. Mayer et al. (1995) define trust in the context of interpersonal relationships, yet the concept may be transferable to the domain of human-AI interaction: “the willingness of a party to be vulnerable to the actions of another party based on the expectation that the other will perform a particular action important to the trustor, irrespective of the ability to monitor or control that other part” (p. 712). Lee and See (2004) further developed Mayer et al.'s concept, defining trust as

“the attitude that an agent will help achieve an individual's goals in a situation characterized by uncertainty and vulnerability” (p. 54). A lack of trust in a system can result in a disuse of a system which could be used more efficiently, while overtrust may cause a misuse and complacency. Thus, trust, especially calibrated trust, plays a pivotal role in the proper use of technology. It is linked to the safe use of AI in the workplace, and it can influence the psychosocial aspects of AI usage.

As anthropomorphic cues have been shown to affect trust (e.g., Hoff and Bashir 2015), it is crucial to consider the impact of anthropomorphic designs on new types of AI agents. There are two interpretations of anthropomorphism: one focuses on the attribution of human-like abilities by humans, while the other refers to design decisions that can evoke such attributions. Therefore, anthropomorphism can be considered both as a dependent variable or outcome, and as an independent variable or operationalization. Epley (2018) emphasizes the first perspective in his brief definition, describing anthropomorphism as “simply perceiving human-like traits in nonhuman agents” (p. 591). Both aspects, however, remain insufficiently explored—particularly with regard to their relationship with trust.

Anthropomorphism has long been a central research topic in Human-Robot Interaction (HRI), with extensive studies on the relationship between anthropomorphism and trust (e.g., Bartneck et al. 2009; Hancock et al. 2011). In a meta-analysis Roesler et al. (2021) conclude that anthropomorphic design in industrial HRI is not universally beneficial and may even reduce trust. In task-oriented settings, human-like features can distract from critical work elements and negatively impact perceived reliability. Unlike

in social HRI, where anthropomorphism supports interaction, in industrial contexts it can hinder performance if not directly task-relevant (Roesler et al. 2021). These findings highlight the importance of context sensitivity and suggest that anthropomorphic design should be applied with caution. With the increasing development of AI-based assistive technologies, such as voice assistants and chatbots without physical embodiment, the concept of anthropomorphism as a design approach has gained significant attention also within the field of Human-Computer Interaction (HCI). This raises the question of whether and to what extent findings from HRI can be transferred to HCI.

Accordingly, this research explores two overarching questions:

1. Which human-like attributes in AI agents elicit perceptions of anthropomorphism?
2. What is the effect of anthropomorphic design on the perceived trustworthiness of AI systems?

To address these questions, we first conducted a scoping review, aiming to explore the intricate relationship between anthropomorphic design and the perceived trustworthiness of non-human agents. Building upon results of the scoping review, we then present two complementary choice-based conjoint analyses examining which human-like attributes in AI agents elicit perceptions of anthropomorphism and how these attributes relate to trust. The research questions posed in this study may also be applicable to other advanced work systems. The present article focuses specifically on AI due to its increasing capabilities and significance. In this context, the term ‘AI’ is to be understood as a broad term encompassing intelligent and extensively automated systems.

2 Scoping review

The scoping review was performed following the PRISMA guidelines for scoping reviews (PRISMA-ScR) (Tricco et al. 2018). First, the research questions were formulated. After developing the search term and deciding on the eligibility criteria, the search was conducted on multiple databases.

2.1 Identifying the research questions

In order to obtain a comprehensive overview of the research field on the interplay between anthropomorphism and trust and to answer the overarching question, the following subordinated research questions (RQ) were identified and served as basis for search strategy and data extraction. RQ 1 to RQ 5 should form the basis for a more thorough assessment and discussion of the results of RQ 6.

- **RQ 1:** How is anthropomorphism defined?
- **RQ 2:** What are the design elements used to manipulate anthropomorphism?
- **RQ 3:** How is anthropomorphism measured?
- **RQ 4:** How is trust defined?
- **RQ 5:** How is trust measured?
- **RQ 6:** What effect does anthropomorphism have on trust?

2.2 Data collection and eligibility criteria

2.2.1 Search strategy

The systematic search was carried out in October 2023 using the databases of ACM, IEEE, Web of Science, Scopus and PSYINDEX, APA PsycInfo, and APA PsycArticles.

The search string for the scoping review was discussed between the authors and refined multiple times. The following search string was identified as retrieving the most complete and precise results:

(anthropomorph OR humanization OR “human-like”) AND (agent OR assistan* OR system OR chatbot OR smart OR intelligent OR virtual OR digital OR cognitive OR “artificial intelligence”) AND (trust* OR reliance OR reliability).*

2.2.2 Inclusion and exclusion criteria

Since this review examines the connection between anthropomorphism and trust, the manipulation of anthropomorphism and the measurement of trust were defined as inclusion criteria. Based on the approach of Hoff and Bashir (2015), we included experimental studies in which the subjects actually interacted with a system. The ‘system’ has not been further specified in order to be comprehensive. Only systems with embodiment such as robots and vehicles were excluded because studies suggest they are cognitively processed differently (e.g., Modliński 2023). Another exclusion criterion was the context of e-commerce and finance, as also argued by Hoff and Bashir (2015). These domains frequently encompass pre-existing concerns regarding risk, security, and privacy, which exert a substantial influence on the users’ perceptions of trust. In such contexts, trust may be more influenced by transactional expectations, financial incentives, and brand reputation than by factors such as interface design or social cues. The scoping review only included articles published in German or English, but made no restriction in regard to the country of implementation of the study.

2.2.3 Study selection and data extraction

The systematic search resulted in 2543 papers. Due to the overlap between the databases 492 duplicates were removed. The abstracts of the remaining 2051 articles were screened by two authors (MR, BK) in parts. To check for inter-rater reliability 100 abstracts were pre-tested in terms of decision flow and eligibility criteria. The remaining articles were divided between the two authors. Inter-rater reliability (Cohen's $\kappa=0.79$) showed a substantial strength of agreement (Landis and Koch 1977). Conflicts were discussed and resolved.

A total of 60 articles were eligible for full-paper screening. All of the papers were reviewed by the same two authors. Again, conflicts were discussed and solved, resulting in 19 articles to be included for further analysis. Figure 1 provides the flow diagram of this scoping review.

The data from the included articles was extracted using MAXQDA 2020 (VERBI Software 2021) and charted in an Excel-file. Extracted data was categorized in bibliographic information, study characteristics, definition, manipulation and measurement of anthropomorphism, definition and measurement of trust and key results on the interplay between anthropomorphism and trust.

2.3 Results and discussion

The articles included in this review were published between 2007 and 2023, with a greater proportion published

since 2020. More than half of the studies were conducted in the United States with a predominantly undergraduate student sample or online participants recruited via crowdsourcing platforms like Amazon Mechanical Turk. Study tasks included mostly abstract interactions like the TNO Trust Task by van Dongen and van Maanen (2006) (e.g., de Visser et al. 2012; Muralidharan et al. 2014). Other studies applied a target identification task in a military setting (Jensen et al. 2020, 2021, 2022). Only two articles placed their research in a workplace context, suggesting that there is a need for more research in realistic settings, especially in working environments. For example, Pak et al. (2012) performed a study with managers who had to make decisions in human resource scenarios. The other workplace-related study focused on a system failure diagnosis of a nuclear reactor (Cockram et al. 2023). The type of system or agent the participants interacted with were mostly described as “automated agent” (e.g., Jensen and Khan 2022, p. 286) or “intelligent decision support system” (Zou et al. 2020, p. 231).

A detailed overview of the study characteristics of the included articles can be found in the appendix (Table 5). The appendix also contains an overview of the results for the following RQs (see Table 6, 7 and 8).

2.3.1 Anthropomorphism (RQ 1 – RQ 3)

Definition (RQ 1) Anthropomorphism was defined by 5 articles according to Epley et al. (2007), as “the tendency to imbue the real or imagined behavior of nonhuman agents with human-like characteristics, motivations, intentions, or emotions” (p. 864). Nonetheless, a substantial number of articles ($n=10$) did not provide a definition, suggesting a notable lack of underlying theoretical foundation. In multiple studies it is described as “the degree to which an agent exhibits human characteristics” (de Visser et al. 2016, p. 331) or simply as “human-likeness” (Jung et al. 2019, p. 1). More complex definitions also include aspects of socialness. For instance, Jensen et al. (2020) describe anthropomorphism as “assignment of human-like qualities to an entity by an observer” (p. 37) and further point to “evaluating another entity’s ‘social potential’” (p. 37) which is adapted from Nowak (2015). A uniform definition and theoretical basis could not be found.

Anthropomorphic design elements (RQ 2) In approximately two-thirds of the articles, anthropomorphism was manipulated using visual characteristics. Two to three levels were usually selected, ranging from low anthropomorphism (mostly an image of a computer, $n=9$) to high anthropomorphism (through the depiction of an avatar, $n=12$). In 9 studies, these were placed in direct comparison with an actual human being, as can be seen in, for exam-

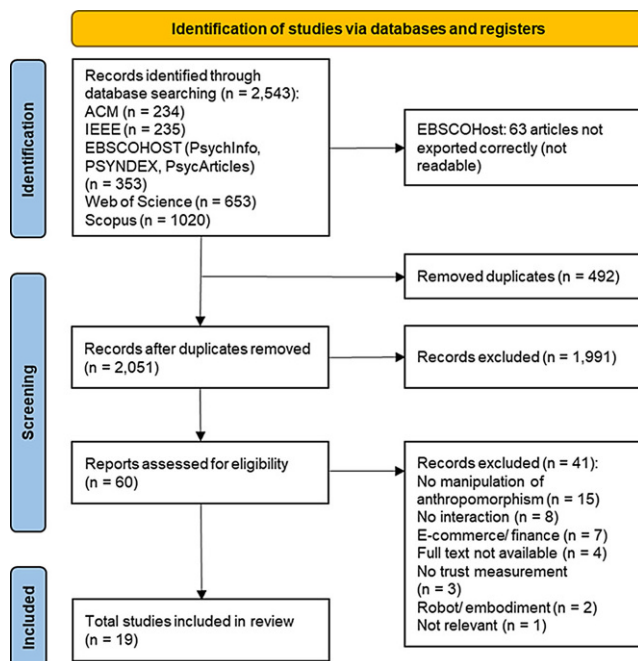


Fig. 1 Flow diagram of scoping review

Abb. 1 Flussdiagramm des Scoping-Reviews

ple, de Visser et al. (2012) or Jensen et al. (2021). While these studies utilize dynamic stimuli (e.g., videos of agents mimic thinking process), others like Smith et al. (2016) use the same stimulus set as static images. The stimuli of de Visser et al. (2012) were adopted and adapted for multiple studies (de Visser et al. 2016; de Visser et al. 2017; Jensen et al. 2021; Jensen and Khan 2022; Smith et al. 2016). On the one hand, this might have led to a bias in the data, but on the other hand, their aim was to ensure a certain degree of standardisation. In addition to their appearance, the agents were given names and brief background stories “to establish an agent’s intent and character” (de Visser et al. 2017, p. 119).

In addition to the many visual manipulations, 4 studies varied the communication style of the agents, for example Sobel and Sims (2022). Here, a distinction was usually made between a machine-like and a human-like style. The machine-like style was represented by short, neutrally formulated sentences. In the human-like style, personal pronouns and direct addresses were used.

Seven studies included the voice of the agent in their research. In short video sequences of the agents from the studies by de Visser et al. (2012) or by Kulms and Kopp (2019), audio-visual utterances by the avatar and humans or the beeping of the computer were implemented as additional stimuli. However, these were not systematically varied or analysed separately. As one of two, Muralidharan et al. (2014) investigated the effects of synthetic speech in cognitive agents ranging from machine-like to human voice characteristics. In addition to this study, Abdulrahman and Richards (2022) also conducted research comparing human and synthetic voices in intelligent virtual voices.

Measurement (RQ 3) A measurement of anthropomorphism was conducted in 12 studies as a manipulation check for the successful humanisation of the agents. Nine studies used the Godspeed questionnaire by Bartneck et al. (2009), an instrument to measure the perception of robots across five indices including anthropomorphism on a 5-point differential scale (for example, “machine-like/human-like”). Additional indices were likeability (e.g., “dislike/like”) and perceived intelligence (e.g., “ignorant/knowledgeable”). In addition to this questionnaire, Jensen et al. (2020) developed three further items to measure perceived behavioural anthropomorphism on a 7-point Likert-scale (e.g., “The system communicated with me like a human would”). Sobel and Sims (2022) included scales of mindful and mindless anthropomorphism from Kim and Sundar (2012) which are designed to capture whether the tendency to treat a computer as human occurs unconsciously (mindless) or consciously (mindful). For the mindful scale three items should be rated on a 10-point differential scale (e.g., “natural/unnatural”).

The mindless scale comprises of four adjectives (e.g., “likeable”) that should be rated on a 10-point Likert-scale.

The manipulation check is essential to verify the successful manipulation of anthropomorphism. Without this verification, it would be inappropriate to assume with certainty that agents with higher degree of anthropomorphic characteristics were actually perceived as more human-like. This was exemplified in the study by Jensen and Khan (2022), wherein the avatar was not perceived as more human-like than the computer. This observation was made for the anthropomorphism index of the Godspeed questionnaire. The Godspeed Questionnaire has been the subject of some degree of criticism. Ho and MacDorman (2010), for instance, have expressed concern regarding the lack of empirical testing and validation of the items. The high correlation between the indices is also viewed critically. Conversely, the perceived anthropomorphism instrument developed by Nowak et al. (2009) showed a significant main effect for the agent’s anthropomorphism. They measure perceived anthropomorphism on three items on a 7-point Likert-scale (e.g., “Does this image look human?”). Both measurements have shown a comparable level of reliability (Godspeed: $\alpha=0.88$; perceived anthropomorphism: $\alpha=0.92$). While Nowak’s items explicitly ask about human-likeness (human look, appearance, expressions), the Godspeed Questionnaire captures anthropomorphism in a more figurative way (“fake/natural”, “unconscious/conscious”). This discrepancy indicates a necessity for a uniform understanding of anthropomorphism and, consequently, the development of valid and precise measurement instruments to quantify anthropomorphism within the domain of HCI. However, this is not within the scope of the present research paper.

2.3.2 Trust (RQ 4 – RQ 5)

Definition (RQ 4) Similar to anthropomorphism, only about half of the articles ($n=11$) include a definition of trust in the introduction of their research. The most prominent definition in seven articles is taken from Lee and See (2004) who define trust as “the attitude that an agent will help achieve an individual’s goals in a situation characterized by uncertainty and vulnerability” (p. 54). This definition is adapted from Mayer et al. (1995), which can also be found in four of the reviewed articles. They define trust as “the willingness of a party to be vulnerable to the actions of another party based on the expectation that the other will perform a particular action important to the trustor, irrespective of the ability to monitor or control that other part” (p. 712). Muralidharan et al. (2014) used the definition of Madhavan and Wiegmann (2007). According to them, trust is “the expectation of, or confidence in, another and is based on the probability that one party attaches to co-operative or

favourable behaviour by other parties” (p. 280). Trust and trustworthiness are sometimes used interchangeably.

Measurement (RQ 5) The different approaches to measure trust need to be differentiated between objective behavioural trust and subjective perceived trustworthiness. As an objective measure of trust, researchers calculate the trust appropriateness (Jensen et al. 2020), take a measure of dependence—number of times participants agreeing or disagreeing with the decision aid or asking for help—(Pak et al. 2012) or interpret the compliance rate as a numeric measure of trust (de Visser et al. 2016). Subjectively perceived trust is measured using questionnaires to obtain subjective assessments of perceived trust or trustworthiness, either as single-items or as scales. Single-items like “What was your trust in the agent?” (e.g., Cockram et al. 2023) are answered on a Likert-scale from low to high trust and were used in 7 studies. The trust in automation scale developed by Jian et al. (1998, 2000) was used twice in studies and comprises of 3 items (e.g., “The system is reliable.”) which are rated on a 5-point Likert scale. The perceived trustworthiness items from Mayer and Davis (1999) were adopted 5 times (e.g., Abdulrahman and Richards 2022). A multitude of questionnaires and measurement methods are used to assess trust and trustworthiness, which is a limitation for the ability to interpret the results. The interchangeable use of trust and trustworthiness also makes it even more difficult to compare the results.

2.3.3 Effect of anthropomorphism on trust (RQ 6)

The results of the included studies present a complex and inconsistent picture. Of the 19 studies included in this review, 8 reported significant effects of anthropomorphism on trust, while 4 found none. Furthermore, 7 studies reported partial or mixed effects.

Positive effects A total of 8 studies demonstrated that anthropomorphism can exert a beneficial influence on the development of trust, particularly with regard to its resilience. Initially, a greater level of trust is placed in computers. However, in the case of trust breaches, the loss of trust is significantly steeper than in the case of avatars or human agents. For instance, de Visser et al. (2016) demonstrate that anthropomorphic agents are more likely to maintain trust following a failure, particularly in conditions of uncertainty. In their third experiment, human-like trust repair behaviors were effective in mitigating the adverse effects of a machine-like appearance. These findings were in line with their earlier (de Visser et al. 2012) and later work (de Visser et al. 2017). Other studies in this group similarly found that anthropomorphic cues enhanced both subjective and behavioral measures of trust (Pak et al. 2012).

In this group, studies were predominantly conducted in controlled laboratory settings ($n=6$) with student samples ($n=5$) and manipulated anthropomorphism as a within-subject design ($n=5$). More than half of the studies involved logical or abstract tasks. Two studies were set in a medical/counseling context. One study conducted a social task, namely a self-disclosure interview. In nearly all studies, the type of anthropomorphic cue was visual, including a short video of the agent. Muralidharan et al. (2014) were the only ones to manipulate the voice. The manipulation check was successful in all five studies that conducted one. System reliability was manipulated in 5 studies and was shown to interact with anthropomorphic features to influence trust outcomes. This assertion is further substantiated by the observation that systems perceived as more reliable were also regarded as more human-like, suggesting a potential bidirectional relationship. Although anthropomorphism helped to maintain trust and mitigate its loss in the event of failure, it did not consistently improve trust calibration, the alignment between trust and the system’s actual performance.

No effect Conversely, 4 studies report no significant effects of anthropomorphism on trust. These null findings often occurred in laboratory settings ($n=3$), between-subject manipulation of anthropomorphism ($n=3$) with student samples in 2 studies. Task domains varied from managerial decision-making (Tuncer and Ramirez 2022), workplace-simulation (Cockram et al. 2023), logical pattern recognition (Zou et al. 2020) to answering common knowledge questions.

The anthropomorphic cues used in this group were often visual and include a name and background story (e.g., Cockram et al. 2023), though their manipulation was only partially successful. One study reported a successful manipulation check for their communication manipulation (Sobel and Sims 2022). The 2 remaining studies did not report a manipulation check.

Mixed results Surprisingly, a large proportion of the studies ($n=7$) showed mixed results and only partial confirmation of the assumption that anthropomorphism has a positive influence on trust. In these cases, anthropomorphism influenced specific dimensions of trust or trustworthiness (e.g., benevolence, warmth) or had effects that were conditional on other variables such as system reliability.

Only 2 studies were conducted in a laboratory setting while the majority was conducted online with a broader non-student sample. Most of the studies varied anthropomorphism between subjects ($n=5$). Task domain varied greatly. The studies from Jensen et al. (2020, 2021, 2022) were set in a military context, while Abdulrahman and Richards (2022) chose a mental counseling task. Kahr et al. (2023) asked the participants to make decisions for legal

cases. The anthropomorphic cue was mostly visual ($n=4$). Two studies manipulated the communication style (e.g., Kahr et al. 2023). The voice was only manipulated by Abdulrahman and Richards (2022). For 3 studies the manipulation check was either partially successful (Jung et al. 2019) or only for the measure of behavioral anthropomorphism (Jensen et al. 2020, 2021). One successful and one failed manipulation check was reported. Two studies didn't measure it.

In some cases, the perception of anthropomorphism did not differ significantly across conditions, and anthropomorphic features did not lead to increases in either perceived or behavioral trust (Jensen and Khan 2022). Notably, anthropomorphism also did not improve trust appropriateness or calibration for Jensen et al. (2021) or Jensen et al. (2020). In some instances, trust-dampening interventions were only effective for agents with machine-like designs, implying that anthropomorphic agents may lead to higher expectations and, consequently, greater resistance to corrective feedback (Jensen and Khan 2022).

Furthermore, they found human-like communication styles to be associated with increased perceptions of benevolence, although not necessarily with higher ratings of ability or integrity. It is interesting to note that the social hormone oxytocin increased trust only for reliable anthropomorphic agents, but not for unreliable ones. Furthermore, anthropomorphic agents still exhibited lower trust resilience compared to actual human agents (de Visser et al. 2017).

Summary Taken together, these findings suggest that anthropomorphism can influence trust, but its effect is neither uniform nor unconditional. Rather, trust responses appear to depend heavily on contextual factors such as task complexity and setting, system reliability and the specific type of anthropomorphic cue. While features like human-like communication or appearance may trigger intuitive social responses that foster trust, these effects may be undermined when agents fail to meet performance expectations. Moreover, the potential for over-reliance on anthropomorphic agents highlights the importance of calibrating trust to system capabilities rather than appearances alone.

3 Conjoint analyses

The scoping review revealed that the most common manipulation in previous studies is an anthropomorphic appearance. Other manipulations included the voice output, a machine-like or human-like name or communication style. However, prior research focused often on individual aspects of an anthropomorphic design in isolation, without systematically comparing their relative impact. As a result, it remains unclear which attributes are (most) influential

in shaping the perception of anthropomorphism and trust. Therefore, we conducted two complementary conjoint analyses to find out which of the manipulated attributes found in the scoping review have the greatest influence on the perception of systems.

Conjoint analyses are a multivariate statistical technique widely applied in marketing research to examine consumer preferences and decision-making processes. They are based on discrete choice experiments and build up on the work of Luce and Tukey (1964). By systematically varying attribute levels and analysing participants' choices, conjoint analyses allow for the quantification of the relative importance of specific attributes and their impact on decision-making and so mimic real-world complex decisions (Orme 2010).

3.1 Research question and hypotheses

The first conjoint analysis will examine the effect on the perceived human-likeness (hereafter referred to as Conjoint Analysis Anthropomorphism (CAA)). The second conjoint analysis will look at the influence of the same attributes and characteristics on the perceived trustworthiness (hereafter referred to as Conjoint Analysis Trustworthiness (CAT)).

The scoping review showed that technical systems are perceived as less human-like than an avatar. There is still no evidence as to whether a computer image is perceived as more human-like than a control condition without an image or a synthetic voice compared to no voice. In the control condition, participants can imagine the agent's appearance or voice, potentially attributing more human-like qualities. In contrast, a synthetic voice or computer image may emphasize the artificial nature of the agent, making it seem less human-like. Therefore, it is assumed that the control conditions are perceived as more human-like than the technical stimulus. Furthermore, there are no analyses yet on which attributes have a greater influence on the perception of human similarity or trustworthiness. This will be explored further following an explorative approach. It is conceivable that more salient attributes, such as appearance, have a greater influence, while less salient attributes, such as name, have the least influence.

We pose the following hypotheses for the CAA:

H1 There is a difference in the perceived human-likeness for the respective attributes in the following directions (from least to most human-like), with no difference between the male and female version:

- a) Name: DX-7385 < System Assistant < Kai/Kira
- b) Appearance: computer < no appearance < human avatar (male/female)
- c) Communication style: objective-machine-like < objective-human-like < extraverted human-like

d) Voice: Synthetic voice < no voice < human voice (male/female)

For the CAT the following hypothesis is postulated:

H2 Human-like attributes will be perceived as more trustworthy in contrast to machine-like attributes.

3.2 Methods

3.2.1 Adaptive choice-Based conjoint analysis

Several methodological approaches exist within conjoint analysis, including rating-based, ranking-based, choice-based, and adaptive conjoint methods. Choice-based conjoint analysis (CBC), the most commonly used approach, presents participants repeatedly with a set of alternatives and requires them to select their preferred option. CBC is a decompositional method in which a stimulus is broken down into its constituent attributes. These attributes are then assigned different levels. During CBC, test subjects repeatedly select their preferred option from various combinations of these levels (choice sets). A weakness of the traditional CBC is the large number of decisions (trade-offs) that have to be made by participants. One possible solution is to enlarge the number of participants. Another would be to utilize an adaptive approach, the adaptive choice-based conjoint analysis (ACBC) to examine decision-making processes more economically. Adaptive conjoint analysis dynamically adjusts attribute levels based on previous responses (Fiedler et al. 2017; Homburg et al. 2022; Orme 2010).

The ACBC approach was chosen to implement this study. It was implemented in the online platform of 1000minds (www.1000minds.com). Here, the PAPRIKA (Potentially All Pairwise Rankings of all Possible Alternatives) method is used to reduce the number of trade-offs (Hansen and Omblér 2008). Using the principle of transitivity, conclusions are drawn from the test subjects' past decisions about possible other combinations, which then no longer need to be explicitly tested.

The outcomes of conjoint analysis include part-worth utilities and the relative importance of attributes and their levels. Part-worth utilities are numerical values that represent the extent to which each attribute level influences respondents' choices. Since these utilities are calculated independently within each attribute, comparisons across different attributes are not directly meaningful. To enable such comparisons, relative importance must be computed. This metric is a normalized value ranging from 0 to 100%, indicating how influential an attribute is in the overall decision-making process. The relative importance is determined based on the difference between the highest and lowest part-

worth utilities within an attribute, with the lowest level serving as the reference point (Orme 2010).

3.2.2 Study procedure

Two independent conjoint analyses were conducted online to assess the perceived human-likeness and trustworthiness of the presented stimuli. Both analyses followed a comparable study procedure, ensuring methodological consistency and allowing for the systematic evaluation across the two constructs. Although the studies do not build on each other, they complement each other. The results of the CAA inform the interpretation of the CAT in terms of the human-likeness of the stimuli.

After giving informed consent, participants were introduced to the study context, which centred on *MAT-Pro-Line*, an AI-based system designed to assist in the (remote) control of a complex production plant. This context is intended to simulate a potential workplace. The task is taken from the Multi Task Attribute Task Battery (MATB) first implemented by Comstock and Arnegard (1992). To ensure a comprehensive understanding of the system, participants were presented with a brief cover story and a short one-minute video demonstrating the OpenMATB (Cegarra et al. 2020) graphical user interface (GUI). A screenshot of OpenMATB is displayed in Fig. 2. Subsequently, participants received detailed instructions on how to complete the conjoint analysis.

Only in the CAT, an additional pre-ranking task was included, requiring participants to rank the levels of the four attributes based on their perceived human-likeness before proceeding to the main task. Participants took part in only one of the two studies.

The core component of the study involved completing a series of trade-off tasks, in which participants were required to make multiple forced-choice decisions concerning either human-likeness (CAA) or trustworthiness of the presented system (CAT). Upon completion of the conjoint analysis, participants provided demographic information (age, gender) and completed the Affinity for Technology Interaction (ATI) scale (Franke et al. 2019). An example of the trade-off and two choice-sets is shown in Fig. 3. As it can be seen, only partial-profiles (two attributes) were applied, because they are less demanding for repeated choices than full-profiles with all attributes alternating. The study was conducted in German. Stimuli and screenshots were translated for publication purposes only.

The experimental procedure was reviewed and approved by the ethics committee of the Federal Institute for Occupational Safety and Health (BAuA). The study was pre-registered on the Open Science Framework platform (osf.io/ywkrp).

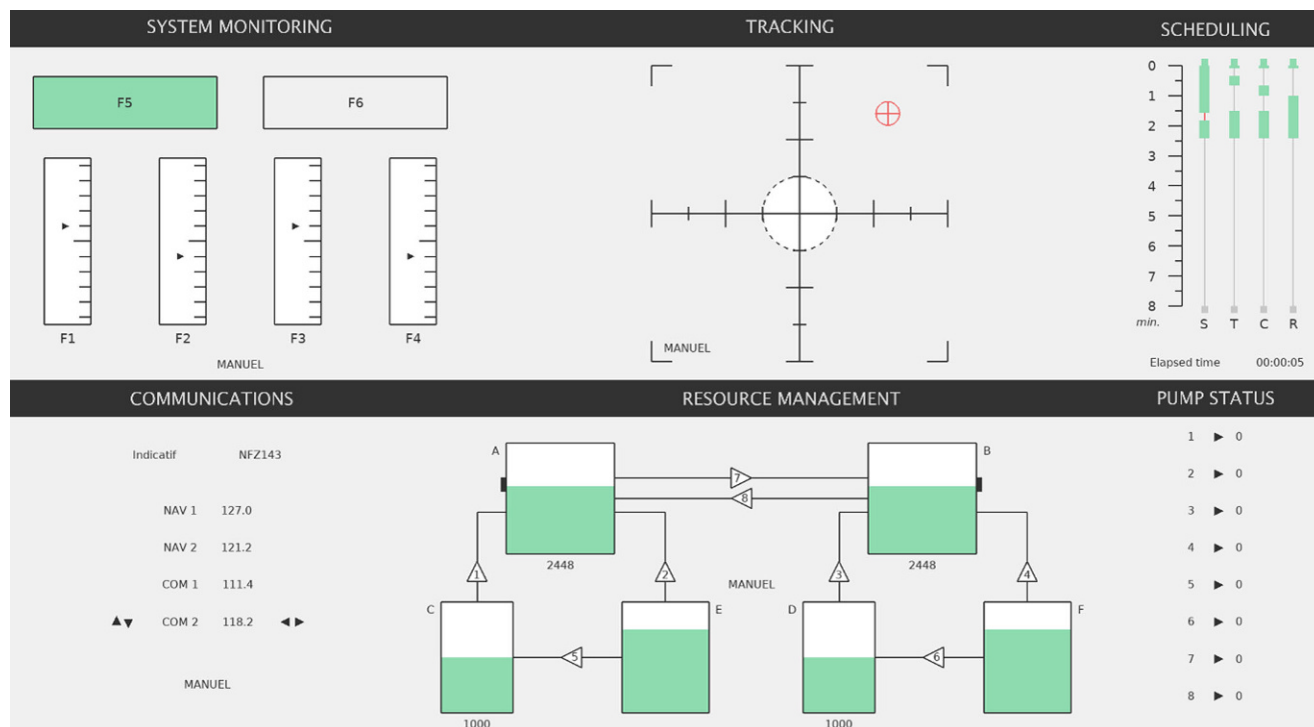


Fig. 2 Screenshot of OpenMATB-GUI

Abb. 2 Screenshot der OpenMATB-Benutzeroberfläche

Fig. 3 Example choice-set: participants had to choose between the two alternatives (translated from German)

Abb. 3 Beispiel Entscheidungs-set: Teilnehmende mussten zwischen einer der beiden AI-Alternativen wählen (aus dem Deutschen übersetzt)

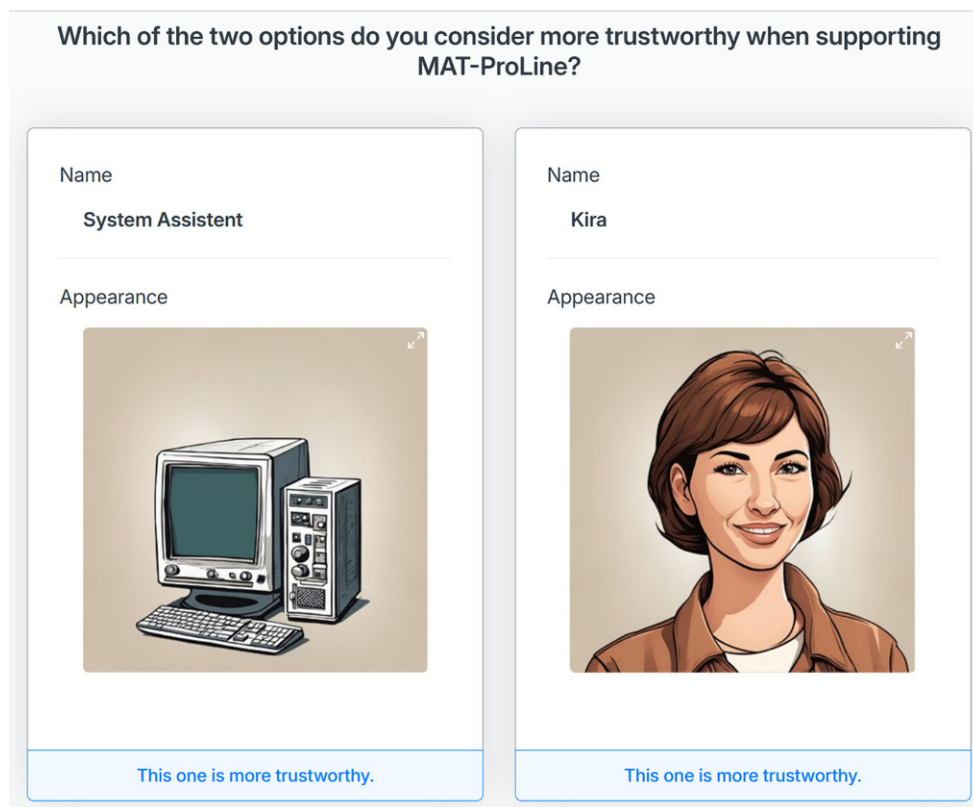




Fig. 4 Stimuli for the attribute appearance: control condition without image (originally labeled: “Kein Bild”), computer, female avatar, male avatar

Abb. 4 Stimuli für das Attribut Aussehen: Kontrollbedingung ohne Bild (ursprünglich mit „Kein Bild“ beschriftet), Computer, weiblicher Avatar, männlicher Avatar

3.2.3 Stimuli

The stimuli used in the study were developed based on insights from the preceding scoping review, which identified key anthropomorphic manipulations applied in prior research. These manipulations were synthesized and abstracted to form a structured set of experimental conditions. The abstracted stimuli were grouped into four attributes: *name*, *voice*, *communication style*, and *appearance*. Each attribute was systematically manipulated along multiple levels.

The attribute *name* included four conditions: DX-7385, System Assistant, Kai, and Kira. The attribute *appearance* comprised four variations: a computer icon, a male avatar, a female avatar, and a no-image control condition, as can be seen in Fig. 4. The pictures were artificially generated with Canva (www.canva.com). The communication style was manipulated across three distinct levels. The participants were shown the example sentences to provide a better idea of the style. An objective-machine-like style was characterized by short sentences without pronouns (“Assistance function activated. Assists when needed.”). The objective-human-like style used pronouns while maintaining factual and neutral language (“I am ready to support you in your task. Let me know if I can help.”). The extraverted-human-like style was mirroring a conversational and engaging interaction style similar to large language model communi-

cation (“I am looking forward to supporting you in your task! Just let me know if you need my help.”). Lastly, the voice attribute was varied across four conditions: a synthetic voice, a male voice, a female voice, and a no-voice control condition. Due to the limitations of the online platform, no actual auditory output was provided. When the study was designed, attention was paid to ensure that no gender-incongruent stimuli appeared in combination, e.g., a male avatar and a female name.

3.3 Sample

Participants were recruited via the Prolific platform and university mailing lists and received a monetary compensation of €7.50 (equivalent to £6.28). No other selectivity bias than that known for (online) studies can be reported. Prolific was selected because it was found to offer high data quality compared to other platforms (Peer et al. 2022). Eligibility criteria included an age range of 18 to 65 years and fluency in German. Sample size determination followed an established heuristic to ensure a minimum of 500 observations per attribute level, applying the formula $n \geq (500 \times c) / (t \times a)$, where c denotes the maximum number of attribute levels, t the number of trade-offs, and a the number of choices per trade-off. Based on a priori simulations using the 1000minds tool, the number of trade-offs was set to 25. A conservative approach was adopted, setting the constant to 1000. Given four attribute levels, 25 trade-offs, and two choices per trade-off, the final sample size requirement was determined as $n \geq (1000 \times 4) / (25 \times 2)$, resulting in a minimum sample size of 80 participants.

3.3.1 Anthropomorphism study

The anthropomorphism study included an initial sample of $N=80$ participants recruited via Prolific. Following data quality checks, four participants were excluded (three due to failed attention checks and one due to technical difficulties), yielding a final sample of 76 valid responses. Of these, 25 participants self-identified as female, 50 as male, and 1 as diverse. The mean age of the sample was 31.97 years ($SD=8.74$). The ATI-score had a mean of 4.48 ($SD=1.00$) with high internal consistency ($\alpha=0.93$). The scale consists of 9 items (like “I like to occupy myself in greater detail with technical systems”) that are answered on a 6-point Likert scale (completely disagree = 1, completely agree = 6) (Franke et al. 2019).

3.3.2 Trustworthiness study

For the trustworthiness study, a total of $N=123$ participants were recruited, comprising 106 via Prolific and an additional 17 through university mailing lists. None of the

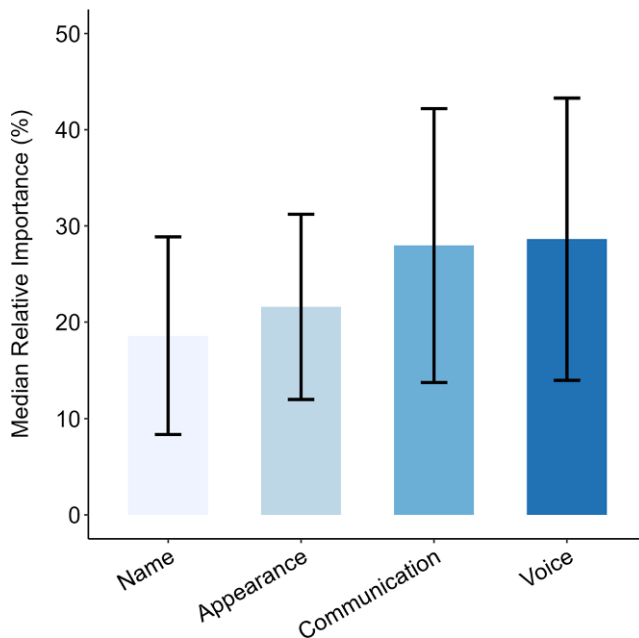


Fig. 5 Median relative importance (in %) for the attributes for perceived human-likeness, error bars indicating IQR

Abb. 5 Median relative Bedeutung (in %) für die Attribute für die wahrgenommene Menschenähnlichkeit, Fehlerbalken zeigen IQR an

participants had previously taken part in the CAA. One participant was excluded due to a failed attention check, resulting in 122 valid responses for analysis. The final sample included 44 female, 77 male, and 1 diverse participant. The mean age was 32.52 years ($SD=10.63$). The ATI score averaged 3.95 ($SD=0.95$), with the scale demonstrating high internal consistency ($\alpha=0.93$).

3.4 Data analysis

Part-worth utilities of the conjoint analyses were computed using the PAPRIKA-method, implemented in 1000minds, and subsequently exported to a CSV file for further processing. For a detailed description of the PAPRIKA-method and calculation of part-worth utilities and relative importance see Hansen and Omblér (2008). The data analysis and visualization were conducted using RStudio (version 4.4.3) with the following packages: dplyr (Wickham et al. 2025a), forcats (Wickham 2023), ggplot2 (Wickham 2016), rstatix (Kassambara 2023), tidyr (Wickham et al. 2025b).

As the assumptions for parametric statistical procedures were not met and heavily violated (data not normally distributed and no homogeneity of variances), non-parametric methods were employed for the analysis. To examine differences between attributes and their respective levels, the Friedman test was applied as a non-parametric alternative to the analysis of variance (ANOVA). The only assumption for the Friedman test is the independency of the samples and the dependency between the repeated measures (in this case

attributes and their levels), which are at least ordinal scaled. This assumption is valid since the relative importance is an interval scale. The significant Friedman tests were followed by multiple post-hoc comparisons which were calculated applying the Wilcoxon signed-rank test for dependent measures. For this, observations with zero differences were removed prior to rank calculation. Standardized z-statistics were obtained by converting the test statistics returned by the Wilcoxon tests into z-scores (Hollander et al. 2013). Since the direction of the contrast can be chosen arbitrarily, absolute z-values are reported. Significance levels were adjusted by the Bonferroni-Holm method to counteract the multiple testing problem. Effect sizes were calculated from the absolute value of the determined z-statistics (Fritz et al. 2012) for significant pairwise comparisons only, as recommended by Field et al. (2012) and interpreted the order of magnitude following Cohen (1992).

The Wilcoxon rank-sum test was applied as the non-parametric alternative for independent samples t-test to test for differences between the male and female sample. For this analysis, the data records of persons who identified themselves as diverse were excluded because there were too few to be statistically evaluated.

The medians and interquartile ranges (IQR) are shown in the following plots, as these are more accurate for non-normally distributed data.

3.5 Results

In the following, first the results of the CAA will be reported, followed by the results of the CAT. Each part reports the descriptive statistics for the attributes, followed by a more in-depth description and inferential analysis of the attributes' level.

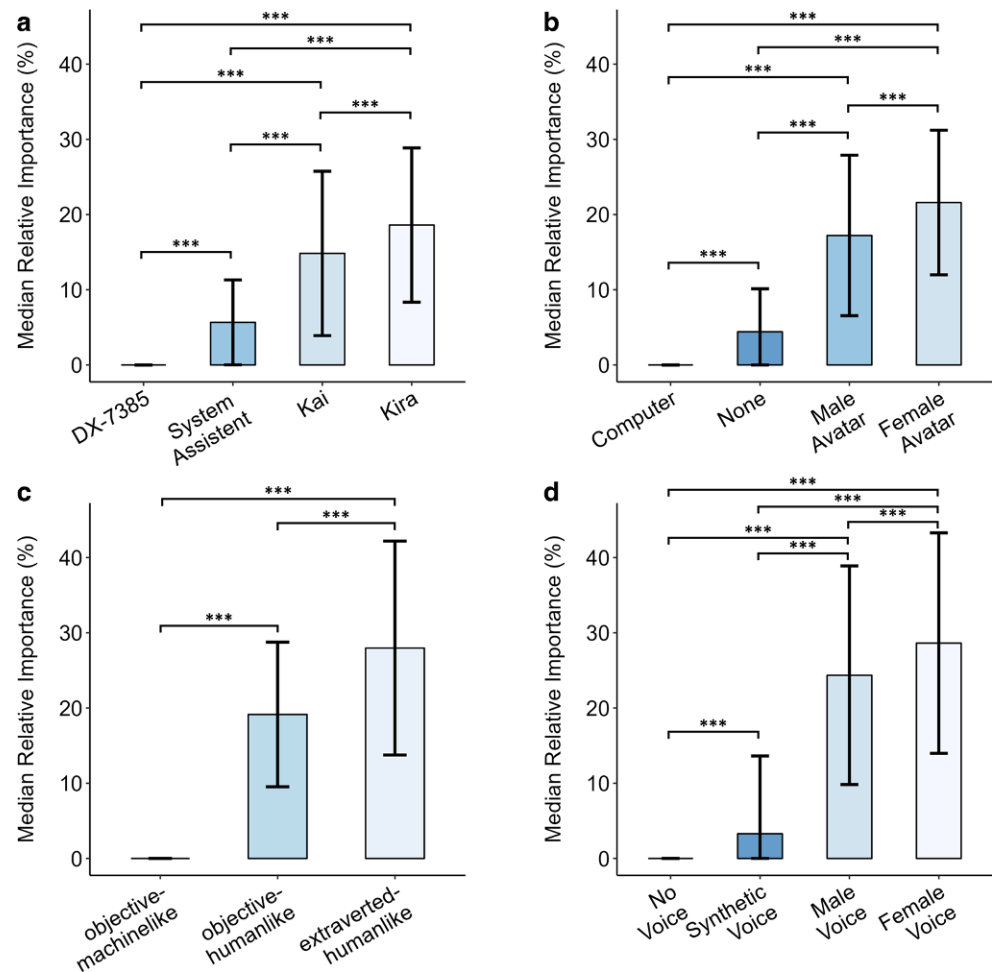
3.5.1 Anthropomorphism study

Attributes In the CAA, 76 participants made between 26 and 40 trade-offs ($M=32.96$, $SD=2.77$). The discrepancy in the number of trade-offs can be attributed to the adaptivity of conjoint analysis. The number of decisions necessary depends on the response behaviour and consistency of the responses.

The attributes with the least relative importance are the attribute name ($M=19.32$, $SD=6.90$, $Mdn=18.61$, $IQR=10.26$) and appearance ($M=23.47$, $SD=9.19$, $Mdn=21.60$, $IQR=9.61$). The communication style is rated second most important ($M=27.90$, $SD=9.80$, $Mdn=27.97$, $IQR=14.22$). The voice is the highest ranked attribute ($M=29.32$, $SD=10.04$, $Mdn=28.63$, $IQR=14.65$). The median and IQR of the relative importance of the attributes are plotted in Fig. 5. No gender difference could be reported for the attributes ($z=0.34$, $p=0.732$, $n=75$).

Fig. 6 Median relative importance (in %) for the perceived human-likeness for the levels of the attributes: **a** name, **b** appearance, **c** communication style, **d** voice, error bars indicating IQR

Abb. 6 Median relative Bedeutung (in %) für die wahrgenommene Menschenähnlichkeit für die Level der Attribute: **a** Name, **b** Aussehen, **c** Kommunikation, **d** Sprache, Fehlerbalken zeigen IQR an



Name (H1a) The levels of the attribute name are ranked lowest for their relative importance in the perception of human likeness. The technical name was ranked the least human-like and therefore set as reference with all values becoming null. The descriptive name was rated as second least human-like ($M = 7.02$, $SD = 5.50$, $Mdn = 5.67$, $IQR = 5.64$). The male name's relative human-likeness ($M = 15.31$, $SD = 6.82$, $Mdn = 14.83$, $IQR = 10.93$) was lower than the female name's ($M = 19.32$, $SD = 6.90$, $Mdn = 18.61$, $IQR = 10.26$), which was most human-like for the participants within the attribute 'name'. Median and IQR of the attribute name can be seen in Fig. 6a.

To estimate the significance of differences between the attributes levels a Friedman test was performed and found to be significant ($\chi^2(3) = 228$, $p < 0.001$). Post-hoc analyses were performed using Wilcoxon signed-rank tests for dependent measures ($n = 76$), which reached significance with large effect sizes ($z = 7.57$, $p < 0.001$, $r = 0.869$). The tests yielded the maximum possible test statistic for all pairwise contrasts, indicating that all non-zero paired differences were in the same direction. Consequently, the standardised test statistics and derived effect sizes were identical for all

contrasts. This pattern reflects a consistently uniform preference in the same direction across all attribute levels. No differences between the female and male sample can be reported ($z = 1.14$, $p = 0.255$, $n = 75$).

Appearance (H1b) Within the attribute appearance, the image of the computer was least human-like and therefore set as reference with values of null. The control condition with no appearance was perceived as less human-like ($M = 6.38$, $SD = 6.01$, $Mdn = 4.42$, $IQR = 5.72$) than the male avatar ($M = 18.98$, $SD = 9.29$, $Mdn = 17.22$, $IQR = 10.67$), which was again less human-like than the female avatar ($M = 23.47$, $SD = 9.19$, $Mdn = 21.60$, $IQR = 9.61$). This is visualised in Fig. 6b.

Comparisons calculated with the Friedman test showed significant differences between the levels of the attribute appearance ($\chi^2(3) = 228$, $p < 0.001$). Once more, all comparisons ($n = 76$) reached significance (see Fig. 6b) with large effect sizes ($z = 7.57$, $p < 0.001$, $r = 0.869$). No significant difference between the male and female sample was found ($z = 0.90$, $p = 0.367$, $n = 75$).

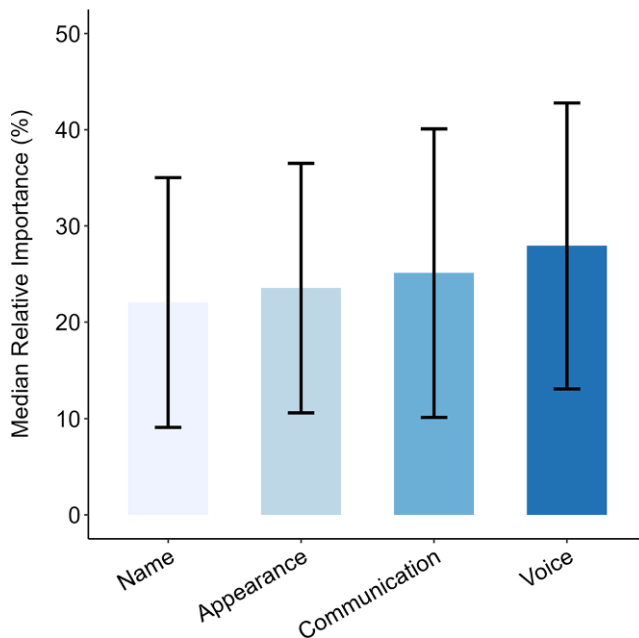


Fig. 7 Median relative importance (in %) for the attributes for perceived trustworthiness, error bars indicating IQR

Abb. 7 Median relative Bedeutung (in %) für die Attribute für die wahrgenommene Vertrauenswürdigkeit, Fehlerbalken zeigen IQR an

Communication style (H1c) The attribute communication style was rated second most important for human-likeness in comparison to the other attributes. The objective machine-like communication was rated as least human-like and set to null-reference. The objective-human-like style was perceived lower in terms of human-likeness ($M=19.05$, $SD=7.78$, $Mdn=19.14$, $IQR=9.62$) compared to the extraverted-human-like communication example ($M=27.90$, $SD=9.80$, $Mdn=27.97$, $IQR=14.22$). The medians and IQR are depicted in Fig. 6c.

The Friedman test conducted to test the differences between the levels of the attribute communication style showed significant differences ($\chi^2(2)=152$, $p<0.001$). As before, all post-hoc contrasts (all $n=76$) between the attribute levels were significant with large effect sizes ($z=7.57$, $p<0.001$, $r=0.869$). These significances are also visualised in Fig. 6c. There was no significant difference between the male and female sample ($z=0.58$, $p=0.561$, $n=75$).

Voice (H1d) The attribute which contributes most to the perception of human-likeness was the voice output of the system. Here, no output was set to null and served as a reference. The synthetic voice ($M=7.56$, $SD=8.35$, $Mdn=3.28$, $IQR=10.34$) was perceived less human-like than the human voices. The male voice ($M=24.34$, $SD=10.67$, $Mdn=24.36$, $IQR=14.53$) was less human-like than the female voice ($M=29.32$, $SD=10.04$, $Mdn=28.63$, $IQR=14.65$),

which showed the highest values over-all. The values of median and IQR are plotted in Fig. 6d.

A non-parametric Friedman test of differences among repeated measures was conducted. The results show a significant difference between the levels ($\chi^2(3)=228$, $p<0.001$). Post-hoc analyses using Wilcoxon signed-rank tests for paired samples ($n=76$) revealed significant differences in all attribute-levels contrasts ($z=7.57$, $p<0.001$, $r=0.869$), as can be seen in Fig. 6d. The identical values across all contrasts are again attributable to the test statistic reaching its theoretical maximum. A Wilcoxon rank-sum test for independent samples showed no significant difference between male and female participants ($z=1.28$, $p=0.200$, $n=75$).

3.5.2 Trustworthiness study

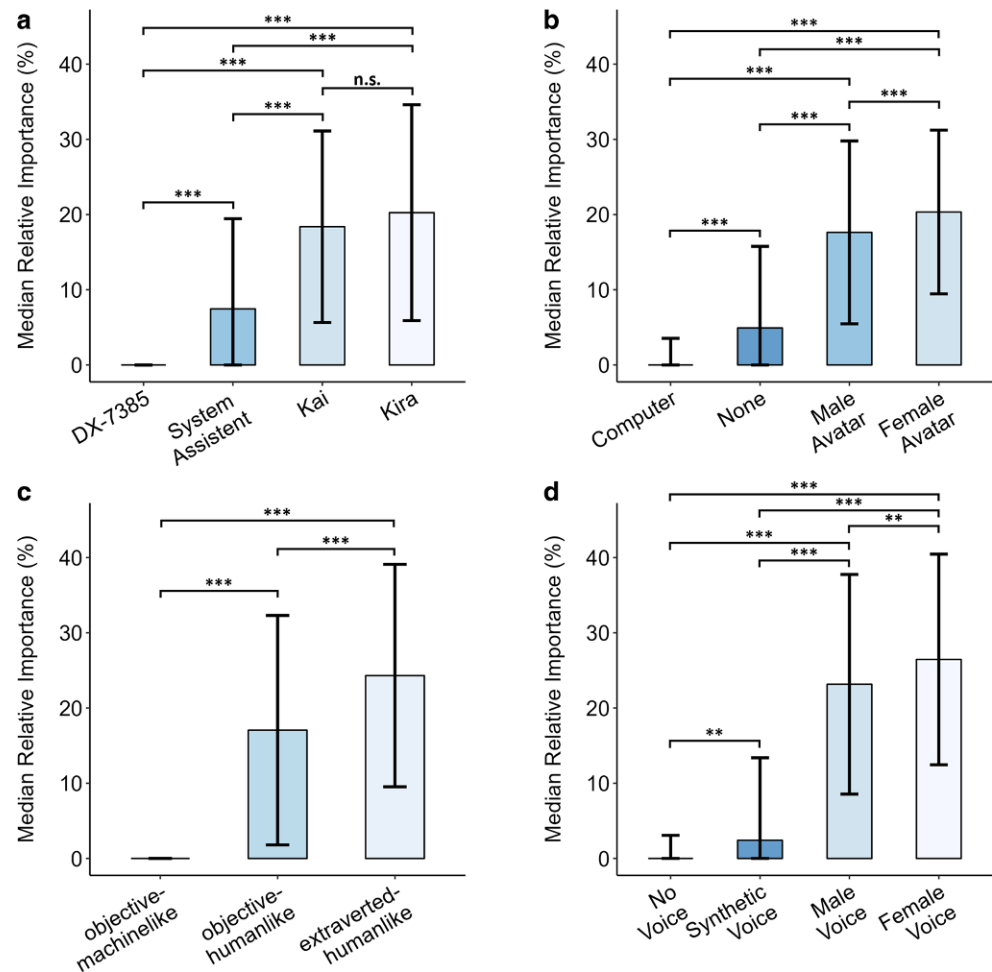
Attributes For the analysis of the CAT, 121 participants were included. They had to make between 26 and 41 trade-offs ($M=33.55$, $SD=2.81$). As participants were asked to pre-rank all attribute levels with respect to their perceived human likeness, the relative importance of an attribute was defined by the highest ranked level for each participant separately which was then averaged over all participants. The relative importance with respect to the perceived trustworthiness of the attribute name ($M=24.28$, $SD=9.39$, $Mdn=22.06$, $IQR=12.96$) and appearance ($M=23.95$, $SD=8.14$, $Mdn=23.55$, $IQR=12.96$) were the lowest. The communication style ranked the second highest relative importance ($M=24.26$, $SD=10.37$, $Mdn=25.11$, $IQR=14.98$). The most important attribute for the perception of a system's trustworthiness was shown by the attribute voice ($M=28.57$, $SD=10.36$, $Mdn=27.94$, $IQR=14.85$). The median relative importance and IQR of the attributes are plotted in Fig. 7. No gender difference could be reported for the attributes ($z=0.34$, $p=0.732$, $n=121$).

Name (H2) Participants rated the technical name as least trustworthy with regard to perceived trustworthiness ($M=3.28$, $SD=7.88$, $Mdn=0.00$, $IQR=0.00$). The descriptive name was rated second least trustworthy ($M=9.22$, $SD=8.66$, $Mdn=7.46$, $IQR=12.00$). Similar results were found for the male ($M=20.95$, $SD=9.96$, $Mdn=18.38$, $IQR=12.73$) and female name ($M=20.72$, $SD=9.89$, $Mdn=20.25$, $IQR=14.35$). Median relative importance and IQR are depicted in Fig. 8a.

The non-parametric Friedman test was calculated to test for significant differences between the levels and revealed a high level of significance ($\chi^2(3)=251.86$, $p<0.001$). When performing the post-hoc contrasts with Wilcoxon signed-rank test (with $n=122$), all became significant ($p<0.001$), as can be seen in Fig. 8a, with medium to large effect size (see Table 1). Only the direct contrast between

Fig. 8 Median relative importance (in %) for the perceived trustworthiness for the levels of the attributes: **a** name, **b** appearance, **c** communication style, **d** voice, error bars indicating IQR

Abb. 8 Median relative Bedeutung (in %) für die wahrgenommene Vertrauenswürdigkeit für die Level der Attribute: **a** Name, **b** Aussehen, **c** Kommunikation, **d** Sprache, Fehlerbalken zeigen IQR an



the male and female name failed to reach significance. No differences between the genders were found ($z=0.63$, $p=0.528$, $n=121$).

Appearance (H2) The relative importance of the level of the attribute appearance were comparable to the attribute name. The image of the computer was perceived as least trustworthy ($M=3.22$, $SD=6.09$, $Mdn=0.00$, $IQR=3.55$). The control condition, where no image was present, were rated slightly higher ($M=7.67$, $SD=8.97$, $Mdn=4.93$, $IQR=10.85$). Differently to the attribute name, the female avatar ($M=21.16$, $SD=8.30$, $Mdn=20.34$, $IQR=10.88$) was perceived more trustworthy than the male avatar ($M=17.92$, $SD=9.88$, $Mdn=17.63$, $IQR=12.16$), which can also be seen in Fig. 8b.

There is a significant difference between the levels of the attribute appearance, as it was shown by the Friedman test ($\chi^2(3)=213.53$, $p<0.001$). Multiple post-hoc comparisons using the Wilcoxon signed-rank test ($n=122$) showed significant results between all levels with moderate to large effect sizes (see Table 2 and Fig. 8b). No significant dif-

ference was found for a comparison between the male and female sample ($z=0.32$, $p=0.748$, $n=121$).

Communication style (H2) The levels of the attribute communication style exhibited a more contrasted pattern. The objective and machine-like communication were perceived as least trustworthy ($M=0.34$, $SD=2.86$, $Mdn=0.00$, $IQR=0.00$). The human-like counterpart was ranked higher ($M=17.88$, $SD=9.54$, $Mdn=17.05$, $IQR=15.25$) but did not match the extraverted-human-like style ($M=22.54$, $SD=11.45$, $Mdn=24.31$, $IQR=14.79$). This pattern can be seen in Fig. 8c.

The Friedman test for differences between the communication styles was significant ($\chi^2(2)=198.75$, $p<0.001$). The post-hoc analysis with Wilcoxon tests ($n=122$) showed differences of significance for all comparisons with large effect sizes (see Table 3). The significances are shown in Fig. 8c. There was no significant difference between the genders for the attribute communication style ($z=0.56$, $p=0.573$, $n=121$).

Table 1 Wilcoxon-comparisons of the attribute name for the CAT**Tab. 1** Wilcoxon-Vergleiche des Attributs Name für die CAT

Comparison	z-value	p-value (adj.)	Effect size (<i>r</i>)
Technical—descriptive	5.02	<0.001***	0.454
Technical—male	8.29	<0.001***	0.751
Technical—female	8.31	<0.001***	0.752
Descriptive—male	9.41	<0.001***	0.852
Descriptive—female	9.20	<0.001***	0.833
Male—female	0.62	0.539	0.056

Table 2 Wilcoxon-comparisons of the attribute appearance for the CAT**Tab. 2** Wilcoxon-Vergleiche des Attributs Aussehen für die CAT

Comparison	z-value	p-value (adj.)	Effect size (<i>r</i>)
No—computer	4.42	<0.001***	0.404
No—male	6.40	<0.001***	0.579
No—female	7.98	<0.001***	0.722
Computer—male	8.61	<0.001***	0.780
Computer—female	9.29	<0.001***	0.841
Male—female	4.22	<0.001***	0.382

Table 3 Wilcoxon-comparisons of the attribute communication style for the CAT**Tab. 3** Wilcoxon-Vergleiche des Attributs Kommunikationsstil für die CAT

Comparison	z-value	p-value (adj.)	Effect size (<i>r</i>)
Machine-like—objective	9.58	<0.001***	0.867
Machine-like—extraverted	9.28	<0.001***	0.840
Objective—extraverted	6.17	<0.001***	0.559

Table 4 Wilcoxon-comparisons of the attribute voice for the CAT**Tab. 4** Wilcoxon-Vergleiche des Attributs Sprache für die CAT

Comparison	z-value	p-value (adj.)	Effect size (<i>r</i>)
No—synthetic	2.62	0.009**	0.237
No—male	9.58	<0.001***	0.868
No—female	9.56	<0.001***	0.866
Synthetic—male	9.44	<0.001***	0.854
Synthetic—female	9.58	<0.001***	0.868
Male—female	3.06	0.004**	0.277

Voice (H2) The least trustworthy was the control condition, where no voice output was described ($M=3.85$, $SD=7.58$, $Mdn=0.00$, $IQR=3.07$). The synthetic voice was ranked slightly higher ($M=6.46$, $SD=8.35$, $Mdn=2.42$, $IQR=10.96$). At a greater distance, the male ($M=24.36$, $SD=10.17$, $Mdn=23.16$, $IQR=14.60$) and female voices ($M=26.69$, $SD=10.59$, $Mdn=26.46$, $IQR=14.00$) ranked close to each other in terms of perceived trustworthiness (see also Fig. 8d).

A non-parametric Friedman test of differences was conducted and rendered a significant result ($\chi^2(3)=282.29$, $p<0.001$). The carried out post-hoc analyses using the Wilcoxon signed-rank test ($n=122$) also showed significant differences for all comparisons (see Table 4). This pattern can also be seen in Fig. 8d. The effect sizes were mostly large. Small to medium effects were found for the contrast control condition and synthetic voice and the contrast between male and female voice. A Wilcoxon rank-sum test for independent samples rendered not significant

for a difference between male and female participants ($z=0.21$, $p=0.834$, $n=121$).

3.6 Discussion conjoint analyses

To answer the research questions of the perception of AI-assistant systems regarding their perceived human likeness and trustworthiness, respectively, two distinct conjoint analyses were conducted. This section will first summarize the results of the CAA and CAT. The hypotheses will be confirmed or rejected. The limitations of the study will be discussed and future research outlined.

3.6.1 Anthropomorphism study

The first conjoint analysis assessed the effect of four attributes on the perceived human-likeness of an AI system and served as a manipulation check of the anthropomorphic manipulations. It was expected, that more salient stimuli, like the appearance of a system, would have the largest

effect on the perception of human-likeness. Furthermore, we hypothesized that there will be a difference in the perceived human-likeness for the attributes' level from technical to human-like stimuli. For the following discussion, we will interpret higher relative importance as being more human-like or more trustworthy, respectively. Furthermore, when interpreting the results, it should be borne in mind that the sample was predominantly young and male, which may lead to bias.

Attributes For the analysis of the perceived human-likeness of a system it can be stated that the attribute name has the least impact, as it would be expected. Surprisingly, the appearance of the system was rated second least. It was hypothesized that the visual stimulus would be of higher salience and therefore more relevant in the perception of human-likeness. On the other hand, this is consistent with the results of the scoping review, which showed that visual manipulation of anthropomorphism was only partially or not at all successful in some studies. Instead, the communication style and voice of the system were considered significantly more important. The results of the attributes are particularly interesting, as previous research has focused primarily on visual manipulations of anthropomorphism, as the review showed. Alongside establishing a name or a brief background story, this is the simplest form of anthropomorphic design. Nevertheless, more complex stimuli appear to be of greater importance for the perception of human-likeness. Communication and voice are crucial factors in human-human interaction and therefore cannot be neglected in human-technology interaction either.

As it was hypothesized, we found most of our predicted rankings of the levels to be accurate. Within each attribute, the levels showed significant differences, indicating overall a successful manipulation. Meaning, more technical stimuli were perceived less human-like than the human-like counterparts.

Name (H1a) For the attribute name, which was ranked lowest, the technical name was perceived as least human-like, as predicted. The descriptive name was less human-like than the human male and female name, as we predicted. What was surprising, was the significant difference between the male and female stimulus. Apparently, the female stimulus is perceived as more human than the male one. When developing the stimuli, care was taken to keep both human names as similar as possible. Therefore, the difference found is likely due to a gender effect, with the female stimulus having an advantage in terms of perceived human similarity.

Appearance (H1b) As predicted, an analysis of the attribute appearance revealed that the computer image was perceived

as less human than the control condition, in which no image was present. The human avatars were perceived most human-like. However, the same difference between the male and female avatar was found as it was the case for the attribute name. The evaluation of a female appearance to be more human-like aligns with prior research by Stroessner and Benitez (2019). They found a female robot to be evaluated as more human-like. It remains to be discussed whether the mere presence of an image makes the system more approachable and tangible, leading to an increased perceived human-likeness.

Communication style (H1c) The communication style was ranked as second most important in the perception of human-likeness in AI systems. Here, the results supported our hypotheses that the objective-machine-like communication style is perceived least human-like compared to the objective-human-like which was less human-like than extraverted-human-like communication. For a more comprehensive analysis, it would have been helpful to include questions about the participants' experience with generative AI language models. This would have been particularly interesting, as the extraverted style had a strong resemblance to such models. For future research it might be of interest to evaluate artificially generated language in terms of its social cues (Feine et al. 2019).

Voice (H1d) We did not expect the attribute voice to be most important for perceived human-likeness. Furthermore, we hypothesized that a synthetic voice would be perceived as less human-like than the no-voice control condition. But apparently, the more modalities (such as voice) are used and the closer they are to human quality, the more human-like they are perceived to be. It may be argued that this arises from ascribing human-like capabilities like communication to the system which in turn leads to higher perceived human-likeness. The human voices were ranked highest, which aligns with our hypotheses. Despite that, the female stimulus was again perceived as most human-like. This is particularly striking, as one would expect everyday exposure to systems like Siri, Alexa or navigation systems, which are equipped with female attributes, would lead to a change in expectations (Luo et al. 2019; Reinkemeier & Gnewuch 2022). This could lead to the perception that female stimuli are less human-like because they are associated with technical devices. Instead, the opposite effect was found.

The attribute voice included the level synthetic voice, which was deliberately non-gendered for reasons of experimental efficiency. In general, only a textual description of the voice output was provided, requiring participants to imagine the auditory characteristics themselves. Some participants reported difficulty with this task and indicated

that an audio example would have been helpful for a more accurate assessment. Nonetheless, several participants still identified voice as the most influential attribute. One participant noted: “I believe the voice is the most important aspect. Even if the name sounds technical or there is no image, the voice is what I engage with most—it leaves the strongest impression on me. The more technical the voice sounds, the less human-like the system appears.”

Gender effects An interesting outcome of the present study is the absence of significant differences between male and female participants in their processing and interpretation of anthropomorphic stimuli. Within the limits of the sample bias, this suggests a generally consistent perception of human-likeness across genders, which stands in contrast to previous findings that have reported gender-specific variations in the evaluation of artificial agents (Eyssel and Hegel 2012; Nass and Moon 2000). Notably, the gender of the stimuli emerged as a significant factor: irrespective of participant gender, systems with female attributes were consistently perceived as more human-like. This aligns with previous findings that associate female agents with traits such as warmth, benevolence, and friendliness (Stroessner and Benitez 2019; Seo et al. 2024). Borau et al. (2021) discussed the interplay between presenting systems as females and the effect of greater human-likeness of such systems. Historically, female voices and personas have been used as the default in navigation systems, customer service bots, and, more recently, voice assistants such as Amazon’s Alexa or Apple’s Siri. This long-standing exposure could lead to a perceptual bias in favour of female agents, enhancing their perceived human-likeness independently of the participant’s own gender.

Further research is needed to disentangle whether this effect arises from perceptual familiarity, stereotypical role associations and expectations. Additionally, it would be valuable to explore whether this female preference holds across different contexts and with a more balanced and diverse sample.

3.6.2 Trustworthiness study

For the CAT, in line with the results of the scoping review, we hypothesized that the more human-like a system is, the more trustworthy it will be perceived. As results have shown, this hypothesis can be confirmed. As previously stated, the sample was predominantly young and male, which may have resulted in bias in the data and should be borne in mind when interpreting the results.

Attributes The clear differences between name, appearance, communication style and voice from the CAA could not be replicated. All attributes contribute rather similar to

the perception of trustworthiness. Nonetheless, the order of attributes was comparable. The attribute name was ranked as least important followed by the appearance of the system. These results are consistent with the scoping review findings, as many studies involving visual anthropomorphism manipulation (and manipulation of name and background story) found no effect on trust, or only a partial one. Like for the CAA, communication style and voice were of the highest relative importance. This also came as a surprise to some of the participants, as one stated after participation: “The communication style had more impact on me than I initially thought. That surprised me a little.”

Name (H2) Taken the results from the CAA: For every attribute the most human-like stimulus was perceived most trustworthy. The human names were the highest ranked when compared to a descriptive name and twice as important for the perception of trustworthiness. The technical name was least important and therefore least trustworthy. There was no observable difference for the male and female name as it has been found for the perceived human-likeness. No known studies have examined the effect of names on trustworthiness separately. This has only been researched in combination with other attributes like appearance or voice. Apparently, the decisive factor seems to be that it is a human name, which outweighs the gender effect. Given that MATB tends to be associated with a male stereotype, it might have been expected that gender stereotypes would be more pronounced. As research from HRI shows, the male name could have been preferred in this case (e.g., Tay et al. 2014; Kraus et al. 2018).

Appearance (H2) The appearance of the system followed the pattern of the CAA. The computer image was perceived as less trustworthy than the control condition that did not include an image. Even so, the participants ranked the human avatars higher. Although the female avatar was perceived as significantly more trustworthy than the male avatar, this effect was moderate compared to the large effect observed for the CAA.

Communication style (H2) The machine-like communication style was rated as least trustworthy. This outcome is unexpected, given that the scoping review revealed that technical and more objective agents (e.g., computers) tend to be trusted more initially. When comparing both human-like expressions, participants rated the extraverted communication as more trustworthy. Despite this significant difference for all participants, there was some feedback indicating a strong rejection for this communication style. One participant stated his disapproval as follows: “Basically, I dislike AI systems that try too hard to appear human. This includes, above all, an overly friendly tone of voice: ‘I’m delighted...’.”

A computer has no feelings, so it cannot be delighted. To me, this comes across as dishonest and ingratiating.”. As discussed earlier, it would have been interesting to evaluate the experience of working with generative language models and their potential pitfalls like hallucinations (Huang et al. 2025). In the context of a potential uncanny valley effect (Mori et al. 2012), it would be intriguing to observe how this evolves with increasing experience involving large language models, avatars, or robots.

Voice (H2) The voice attribute was the most relevant for the perception of trustworthiness. The presence of a voice was found to have a positive impact on the perception of trustworthiness. However, this constituted a comparatively negligible effect. The distinction between synthetic and human voices was one of the most pronounced, indicating that users are sensitive to the naturalness and authenticity of auditory cues when assessing the trustworthiness of artificial systems. The comparison between the male and female voice was significant, showing a small-to-medium effect size. Consistent with the previous results, the female stimulus was perceived as more trustworthy. The positive attitude towards female voice assistants was found before in research (Goodman and Mayhorn 2023). This contrasts with earlier research that has pointed to a gender match effect—where participants respond more positively to agents matching their own gender (Guadagno et al. 2007; ter Stal et al. 2020; Byrne et al. 1967). Furthermore, according to Tolmeijer et al. (2021) there is a great deficiency of research on perceptions of gendered and gender-ambiguous voices.

Gender effects At this point, it is important to note that only binary manipulation of gender was used. The decision was taken not to consider a gender-neutral stimulus in order to ensure that the number of trade-offs in the conjoint analysis was kept manageable. This aspect warrants further investigation in future studies. It is evident that other aspects of diversity (e.g., age, ethnicity, cultural background) were similarly neglected due to the aforementioned reason, and it is recommended that these be explored in future research as well.

Taking all together, we see a preference of human-like stimuli in comparison to functional and machine-like alternatives in terms of their perceived trustworthiness. Furthermore, a significant preference for the female form was found for human attribute levels, as before for perceived human-likeness. There is a distinct tendency for female stimuli to be perceived as more trustworthy, as there is a female advantage in terms of benevolence (Reinkemeier & Gnewuch 2022). Furthermore, it is striking that no gender stereotypes appear to have been activated, as the control activity in the production context corresponds more to a male-dominated job. This bias has already been identified

in work with robots (Bryant et al. 2020; Tay et al. 2014). Future studies could examine more closely the significance of stereotypes in the context of anthropomorphic on-screen agents in the workplace.

A sociotechnical perspective From a sociotechnical systems perspective (Trist and Bamforth 1951), these findings carry significant implications for the domains of workplace safety and well-being. Trust in anthropomorphic agents emerges at the intersection of the technical subsystem (design of the agent, its reliability, transparency) and the social subsystem (user expectations, team norms, organizational culture). While trust in on-screen agents can reduce uncertainty and cognitive effort, it also carries the risk of overtrust and subsequent automation misuse, where workers might follow system recommendations without sufficient scrutiny (Lee and See 2004; Dzindolet et al. 2003). Conversely, distrust in digital agents can lead to resistance, an increased workload, thereby heightening stress and undermining psychological well-being (Ayyagari et al. 2011; Riedl 2012).

Future research should explore how contextual factors—such as task complexity, user expectations, or stress levels—influence responses to anthropomorphic design. Longitudinal and interactive studies are needed to better understand the long-term and situational dynamics of anthropomorphism and trust in workplace settings from a sociotechnical perspective.

3.6.3 Limitations and implications for future research

Although the study provides valuable insights, some questions remain unanswered. Some limitations should therefore be discussed and possible prospects for future research outlined.

As with any online study, the research is subject to certain methodological constraints inherent to this format. While Prolific was chosen as the platform due to its reputation and large participant pool—including a sizable number of German participants—it remains an online, self-administered environment. This can result in reduced control over contextual variables such as participant environment or distraction level. To mitigate this, two attention checks were included at the beginning and end of the study to ensure participant engagement. The low failure rate of these checks supports an acceptable level of data quality.

Another limitation of the study design lies in the absence of direct rating scales for perceived human-likeness and trustworthiness. Instead, participants were asked to assess the contribution of specific factors by choosing between two alternatives. While this approach arguably facilitates more intuitive judgments—particularly in the context of conjoint analyses where participants selected the most human-like

or trustworthy alternative—it does not allow for absolute quantification or inter-individual comparability of trust assessments. Future research could combine comparative and absolute rating methods to triangulate participants' perceptions more comprehensively. Nevertheless, this approach provided a comparative and systematic understanding of the importance of anthropomorphic cues without assigning them an absolute value. As the scoping review demonstrated, previous studies predominantly evaluated anthropomorphism within an attribute rather than between attributes. The results of the present conjoint analyses could therefore also explain why the review revealed an ambiguous study situation. However, these results should only be used as a basis for further research to develop a uniform understanding, definition and conception of anthropomorphism, and to design a valid questionnaire or other measurement method.

In this context, it is also important to note that the systems were only rated comparison wise; there was no actual interaction with the system. Participants were instructed to imagine a work-related scenario rather than engage in actual interactions with the system. While this approach enabled a controlled comparison across conditions, it reduces ecological validity. Real-time, embodied interaction scenarios would likely yield deeper insights into trust dynamics in human-agent communication and should be strived for in future research.

When considering the stimuli, due to the limitations of the software utilized, the study only implemented static stimuli for all attributes. This limits ecological validity, as real-world interactions typically involve dynamic agents. Future studies could incorporate dynamic stimuli (e.g., moving avatar or actual voice output) to assess the impact of motion and interactivity on perceptions of human-likeness and trustworthiness. Furthermore, the study did not include a measure or control for physical attractiveness of the agents, despite prior research indicating its relevance in social evaluations of artificial entities (Khan and Sutcliffe 2013; for a review see Rheu et al. 2020). This omission could confound interpretations of perceived human-likeness or trustworthiness. Including attractiveness ratings in future designs would allow for a more nuanced understanding of its interaction with anthropomorphic cues.

Although gender congruency between the attributes was controlled (e.g., only female name with female appearance), some combinations may have elicited eeriness, potentially invoking an uncanny valley effect (Mori et al. 2012). While this was not the primary focus of the present study, it warrants further investigation, particularly in light of findings related to perceived human-likeness. Furthermore, the current study did not consider additive effects of multiple anthropomorphic cues. It remains an open question whether a higher number of such cues correlates lin-

early with increased human-likeness or trustworthiness, or whether threshold or interaction effects occur. Additionally, although conjoint analysis allowed for the identification of combinations perceived as more trustworthy or human-like, a detailed analysis of which specific combinations led to those judgments was not conducted for this paper. Moreover, more extensive analyses could be carried out to determine the effect of affinity for technology interaction or age in order to control for individual differences among participants. When speaking about the participants, one has to state that the sample was predominantly male, introducing a gender imbalance that may bias the results—especially in light of potential gender-related differences in perception and evaluation of social agents. Future research should aim for a more gender-balanced and diverse sample to validate the generalizability of the findings.

Despite its constraints, this study provides a deeper and more detailed understanding of anthropomorphism. The decompositional approach revealed that certain anthropomorphic attributes exert a stronger influence on perceptions of human-likeness and trustworthiness than others. Furthermore, the systematic comparison enables direct conclusions to be drawn about the relationship between human-likeness and trustworthiness that were neglected in previous research.

4 General conclusions and outlook

Anthropomorphism is a complex, multifaceted construct. This paper provides a comprehensive overview of the current state of research on anthropomorphism and trust by combining insights from a scoping review with empirical findings from two conjoint analyses. Collectively, these methodologies provide a novel perspective on the influence of anthropomorphic design on the perceived human-likeness and trustworthiness of AI systems, particularly within a work-related environment.

The scoping review revealed key gaps in the literature: a lack of theoretical grounding, inconsistent definitions of trust and anthropomorphism, and limited applicability of findings due to varied manipulation strategies and application contexts. Notably, there is little research on anthropomorphism in workplace scenarios, where trust is critical. It can be concluded that the definition of anthropomorphism from Epley et al. (2007) is both the most widely used and the most accurate. The same applies to the definition by Lee and See (2004), which retains the core idea of Mayer et al. (1995) while adapting it to the automation context. The anthropomorphism-index of the Godspeed-questionnaire by Bartneck et al. (2009) proved to be the standard measurement method, although it is not without controversy and still requires validation for the research of HCI. Neverthe-

less, future research should develop a questionnaire that captures the various facets of anthropomorphism and interprets them explicitly for HCI. In order to measure trust, the combination of objective (behavioral trust) and subjective measurement methods (e.g., single-item questions) is the most appropriate approach for identifying discrepancies between perception and behavior.

The insights of the review subsequently informed the design of the two conjoint analyses, which were designed to validate findings from the literature and assess the relative importance of different anthropomorphic design cues on, firstly, perceived human-likeness and, secondly, trustworthiness. The first identified communication style and voice as the most influential factors for perceived human-likeness—outweighing commonly used visual cues like name and appearance. These results align with the Computers Are Social Actors (CASA) paradigm by Nass and Moon (2000), emphasizing the role of interactional features in social perception.

The second analysis replicated this pattern for trustworthiness: human-like agents, especially those with natural voice characteristics, were rated as more trustworthy. Gender effects were less pronounced, and no significant differences were observed between male and female participants—suggesting that general anthropomorphic cues may override gender-specific preferences in non-interactive work settings.

From a theoretical perspective, this study underscores the need to move beyond superficial visual manipulations and focus on more dynamic, communicative features when studying anthropomorphism. Practically, it highlights the importance of context-sensitive design in workplace AI systems. While anthropomorphic cues can enhance trust, their appropriateness may vary across domains. As one participant noted, overly human-like systems can feel intrusive: “Personally, I would feel uncomfortable if an assistive AI came across as human”.

Within a sociotechnical paradigm, the concept of anthropomorphic design does not inherently contribute to the enhancement of workplace health and safety. By presenting agents as more human-like, and thus potentially more trustworthy, there is a risk of fostering a false sense of reliability, potentially overshadowing system limitations and errors (Hancock et al. 2011). In digitalized work contexts, the challenge lies in calibrating trust with the prevention of complacency, in order to assure both safety and long-term well-being.

5 Appendix

Table 5 Bibliographic Information and Study Characteristics of Included Articles
Tab. 5 Bibliografische Informationen und Studienmerkmale der eingeschlossenen Artikel

Source (Author, Year)	Country	Design (Anthropomorphism manipulated within/between)	Sample (incl. User characteristics)	System	Task	Context	Additional Factors
Abdulrahman and Richards (2022)	Switzerland	Laboratory (between)	118 undergraduate students ($M = 20.64$ years; 38 males, 80 females)	Intelligent virtual agent	Mental counseling (study related stress)	(Medical/Mental) Counseling	Co-presence, eeriness
Cockram et al. (2023)	Australia	Laboratory (between)	120 undergraduate students ($M = 20.63$ years, $SD = 6.13$)	Automated agent	Signal detection task (Sorkin et al. 1991, Yamani and McCauley 2016)	Workplace	Reliability (93%)
de Visser et al. (2012)	USA	Laboratory (within)	14 undergraduate students ($M = 23.1$ years, $SD = 2.7$; 10 males, 4 females)	Cognitive agent	TNO Trust Task (van Dongen and van Maanen 2006)	Logical	Reliability (decreased: 100%, 75%, 50% 0%)
de Visser et al. (2016)	USA	Laboratory (3 experiments) (within)	Exp. 1: 20 undergraduate students ($M = 22$, 18–29 years, 40% female); Exp. 2: 17 undergraduate students ($M = 20.4$, 18–26 years, 47% female); Exp. 3: 20 undergraduate students ($M = 20.7$, 18–25 years, 50% female)	Automated agent	TNO Trust Task (de Visser et al. 2012, van Dongen and van Maanen 2013)	Logical	Reliability (decreased: 100%, 67%, 50%, 0%); feedback; familiarity
de Visser et al. (2017)	USA	Laboratory (within)	84 college students ($M = 24.2$ years, $SD = 2.7$, all males)	Automated agent	TNO Trust Task (de Visser et al. 2012, van Dongen and van Maanen 2013)	Logical	Reliability (decreased: 100%, 50%)
Gong and Nass (2007)	USA	Laboratory (2 experiments, Exp. 2 excluded for review) (within)	Exp. 1: 80 undergraduate students (40 males, 40 females)	(talking-face) agent	Exp. 1: self-disclosure interview	Social	Consistency
Jensen et al. (2020)	USA	Online (between)	130 participants ($M = 37.2$ years; 71 males, 59 females)	Automated agent	Target identification task	Military	Reliability (low, high)
Jensen et al. (2021)	USA	Online (between)	264 participants ($M = 37.2$ years; 161 males, 101 females, 2 other)	Automated agent	Target identification task	Military	Reliability (low, high)
Jensen and Khan (2022)	USA	Online (between)	225 participants ($M = 36.9$ years, 145 males, 79 females, 1 other)	Automated agent	Target identification task	Military	Reliability (low, high); trust dampening/apologizing
Jung et al. (2019)	Korea	Laboratory (within)	15 subjects ($M = 25.1$, 21–34 years; 9 males, 6 females)	Agent	One-armed bandit task (modified from Behrens et al. 2007)	Logical	–
Kahr et al. (2023)	Netherlands	Online (between)	171 british participants ($M = 36.6$ years, 71% female)	AI system	Decision-making task estimating jail time for 20 legal cases	Legal/judicial	Accuracy (low, high)
Kulms and Kopp (2019)	Germany	Online (between)	114 participants ($M = 30.49$, 18–54 years, 41.2% female)	Agent	2-player cooperative game	Logical	Advice Quality (low, high)

Table 5 (Continued)
Tab. 5 (Fortsetzung)

Source (Author, Year)	Country	Design (Anthropomorphism manipulated within/between)	Sample (incl. User characteristics)	System	Task	Context	Additional Factors
Muralidharan et al. (2014)	USA	Exp. 1: Online (excluded for review) Exp. 2: Laboratory (within)	Exp. 2: 23 participants (9 males, 14 females 18–28 years)	Cognitive agent	Exp. 2: TNO Trust Task (de Visser et al. 2012, van Dongen and van Maanen 2013)	Logical	Accuracy (decreased: 100%, 50%)
Pak et al. (2012)	USA	Laboratory (between)	45 undergraduate students ($M = 19.82$, 18–26 years, 20 females, 25 males) (older group excluded for review)	Decision aid	Decision making (diabetes management)	Medical	Reliability (67%)
Park et al. (2023)	USA	Online (between)	385 participants ($M = 37.5$ years; 233 males, 152 females)	AI chatbot	Two-way conversation with a mental health AI chatbot counselor	(Medical/Mental) Counseling	Psychological distance
Smith et al. (2016)	USA	Online (within)	105 participants ($M = 34.38$ years; 44 males)	Agent	Reading Mind in the Eyes (Baron-Cohen et al. 2001); Mathematical operations	Social, Logical	–
Sobel and Sims (2022)	USA	Laboratory (between)	128 undergraduate students ($M = 19.05$, 18–30 years; 64 males, 64 females)	(simple text-based) Interface	Answering questions	Common knowledge test	Reliability (80%)
Tuncer and Ramirez (2022)	Canada	Online (between)	134 (managerial) decision-makers	AI agent	Decision-making scenarios	Real-life	Trust facets
Zou et al. (2020)	China	Laboratory (within)	40 participants ($M = 20.0$, 18–25 years; 14 males, 24 females)	Intelligent decision support system	TNO Trust Task (de Visser et al. 2016)	Logical	Reliability (95%, 70%, 45%)

Table 6 Definitions, Manipulations and Measurements of Anthropomorphism
Tab. 6 Definitionen, Manipulationen und Messinstrumente von Anthropomorphismus

Source (Author, Year)	Definition Anthropomorphism	Reference	Manipulation Anthropomorphism	Morphology Anthropomorphism	Measurement Anthropomorphism	Manipulation Check
Abdulrahman and Richards (2022)	“characteristics should be as real as those of human beings”	–	Voice: human, text-to-speech	Voice	Godspeed questionnaire – Likability (Bartneck et al. 2009); voice eeriness (Ho and MacDorman 2010)	Not successful
Cockram et al. (2023)	“the extent to which an automated agent possesses human-like features or characteristics” (Epley et al. 2007)	Epley et al. (2007)	Appearance: computer, avatar, human; names; background stories	Visual + name + background story	Godspeed questionnaire – Anthropomorphism (1 item), likability (3 items), perceived intelligence (3 items) (Bartneck et al. 2009)	Partially successful: no differences for anthropomorphism and perceived intelligence, only for likability (computer < avatar = human)
de Visser et al. (2012)	“humanness of the agent”	–	Appearance: agent videos (computer, avatar, human “thinking”); audio-visual utterances; agent stories	Visual (videos) + background story + voice	Godspeed questionnaire – Machine/Human (Bartneck et al. 2009)	Successful (computer < avatar < human)
de Visser et al. (2016)	“the degree to which an agent exhibits human characteristics”	–	Appearance: computer (blinking lights, GIF-images, series of beeps); avatar (standard avatar agent, online created, “thinking” phrases like human); human (male actor, making sounds, saying phrases)	Exp. 1–3: (audio-) visual (videos) + background story Exp. 3: behavioral (apologizing)	Godspeed questionnaire – Anthropomorphism, likability, perceived intelligence (Bartneck et al. 2009)	Successful (computer < avatar < human)
de Visser et al. (2017)	“ascribe intentionality to inanimate agents” (Epley et al. 2007)	Epley et al. (2007)	Appearance: agent videos (computer, avatar, human “thinking”); audio-visual utterances	Visual	Godspeed questionnaire – Anthropomorphism, likability, perceived intelligence (Bartneck et al. 2009)	Successful (computer < avatar < human)
Gong and Nass (2007)	“But, humanoid, which is defined by this research as being anthropomorphic but clearly artificial and synthetic, is likely a continuum, if not a multidimensional construct.”	–	Appearance: humanoid, human; Voice: humanoid	Visual (video) + voice	Manipulation check (“computer synthesized” or “of a real person”; consistency)	Successful

Table 6 (Continued)
Tab. 6 (Fortsetzung)

Source (Author, Year)	Definition Anthropomorphism	Reference	Manipulation Anthropomorphism	Morphology Anthropomorphism	Measurement Anthropomorphism	Manipulation Check
Jensen et al. (2020)	“Anthropomorphic is used to describe a target that is perceived as humanlike, and more or less anthropomorphic represents the degree to which a target is perceived as humanlike. Degrees of anthropomorphism are synonymous with degrees of socialness, as anthropomorphizing has been referred to as the act of evaluating another entity’s “social potential” and informing expectations of that entity’s future behavior” (Nowak 2015)	Nowak (2015)	<i>Communication</i> : machine-like (minimal, informational, impersonal), humanlike (friendly, apologetic, and framed from the perspective of the automation (automation referred to itself as “I”))	Communication style	Godspeed questionnaire—Anthropomorphism (Bartneck et al. 2009); perceived behavioral anthropomorphism	Successful: only for perceived behavioral anthropomorphism (machine-like < human-like); not successful for Godspeed questionnaire
Jensen et al. (2021)	“the assignment of humanlike qualities to a non-human entity” (Epley et al. 2007)	Epley et al. (2007)	<i>Appearance</i> : agent videos (computer: blinking; avatar: animated character; human: actor is shown thinking about a task)	Visual	Godspeed questionnaire—Anthropomorphism (Bartneck et al. 2009)	Successful (computer < avatar < human)
Jensen and Khan (2022)	“humanlike appearance and communication style”	–	<i>Appearance</i> : avatar, computer; message (apology)	Visual + behavioral (communication style)	Godspeed questionnaire—Anthropomorphism (Bartneck et al. 2009); perceived anthropomorphism (Nowak et al. 2009)	Successful: only for perceived behavioral anthropomorphism (machine-like < human-like); not successful for Godspeed questionnaire
Jung et al. (2019)	“machine human-likeness”	–	<i>Appearance</i> : human, robot face (video); <i>Voice</i> : female, beeping; <i>Internal</i> : personality (risk-taking behavior)	Visual (video) + voice + personality	Human-likeness questionnaire (not specified)	Partially successful (only for external human-likeness (appearance, voice))
Kahr et al. (2023)	“human-like”	–	<i>Communication</i> : humanlike vs. abstract explanation; different names	Communication style	–	–
Kulms and Kopp (2019)	“mimic human appearance and behavior”; “humanness characteristics”	–	<i>Appearance</i> : computer (symbolic icon), virtual (avatar), human (male person); name; introduction	Visual (video) + voice	–	–

Table 6 (Continued)
Tab. 6 (Fortsetzung)

Source (Author, Year)	Definition Anthropomorphism	Reference	Manipulation Anthropomorphism	Morphology Anthropomorphism	Measurement Anthropomorphism	Manipulation Check
Muralidharan et al. (2014)	“human-like”	–	<i>Voice</i> : machine, android, human (pitch contour, flanging)	Voice	Godspeed questionnaire (Bartneck et al. 2009)	Successful (machine < android < human)
Pak et al. (2012)	“users, often unknowingly, treat their computers as social beings, attributing human-like characteristics to them” (Nass et al. 1993); “Interface anthropomorphism occurs when computer automation is given human-like traits and characteristics” (Gong and Nass 2007)	Nass et al. (1993); Gong and Nass (2007)	<i>Appearance</i> : anthropomorphic (picture of static smiling female doctor), non-anthropomorphic (image of gears)	Visual	–	–
Park et al. (2023)	“Individuals tend to perceive human-like characteristics in nonhuman agents and the level of perceived humanness affects psychological, attitudinal, and behavioral outcomes”	Epley et al. (2008); Go and Sundar (2019)	<i>Appearance</i> : human-like vs. machine-like representation of chatbot (with facial expressions); communication	Visual	–	–
Smith et al. (2016)	“humanness”; “human-like”	–	<i>Appearance</i> : computer, avatar, human; background story	Visual + background story	–	–
Sobel and Sims (2022)	“When humans perceive or attribute humanlike qualities to something that is not human, this is known as anthropomorphism”; “mindful anthropomorphism is considered the genuine belief that a nonhuman possesses a humanlike mind, while mindless anthropomorphism is the behavior of treating an entity as if it warrants a humanlike response”	Epley et al. (2007); Araujo (2018); Kim and Sundar (2012); Lee (2010); Nass and Lee (2001)	<i>Communication</i> : humanlike vs machine-like language output	Communication style	Mindful & Mindless Anthropomorphism Scale (Kim and Sundar 2012)	Successful for both measures (non-anthropomorphic < anthropomorphic)
Tuncer and Ramirez (2022)	“Anthropomorphic: The AI agent design imitates some parts of the human anatomy” (compared to humanoid and non-humanoid)	Natarajan and Gombolay (2020)	<i>Appearance</i> : AI agent (AI face generator: selected to be neutral regarding nationality, gender and age) vs. Dashboard	Visual	–	–
Zou et al. (2020)	“Anthropomorphism refers to the attributes that make nonhuman objects such as robots, computers, and animals to be like human, in appearance, voices, behaviors”	Bartneck et al. (2009)	<i>Appearance</i> : avatar vs. computer; names	Visual + names	–	–

Table 7 Definitions and Measurements of Trust
Tab. 7 Definitionen und Messinstrumente von Vertrauen

Source (Author, Year)	Definition Trust	Reference	Measurement Trust
Abdulrahman and Richards (2022)	–	–	<i>Subjective</i> perceived trustworthiness (adapted from Mayer and Davis 1999); (short version) working alliance inventory (Hatcher and Gillaspay 2006)
Cockram et al. (2023)	“the degree to which an agent believes another will help them in achieving their goals” (Lee and See 2004)	Lee and See (2004)	<i>Subjective</i> : single item trust (de Visser et al. 2016; de Vries et al. 2003; Lee and Moray 1992; Lewandowsky et al. 2000)
de Visser et al. (2012)	–	–	<i>Objective</i> : behavioral trust (compliance); <i>Subjective</i> : trust measure (Jian et al. 2000)
de Visser et al. (2016)	“the attitude that an agent will help achieve an individual’s goals in a situation characterized by uncertainty and vulnerability” (Lee and See 2004)	Lee and See (2004)	<i>Objective</i> : behavioral trust (compliance), team performance; <i>Subjective</i> : 10-item Likert scale (de Vries et al. 2003; Lee and Moray 1992; Lewandowsky et al. 2000)
de Visser et al. (2017)	–	–	<i>Objective</i> : behavioral trust (compliance); <i>Subjective</i> : single item trust (de Vries et al. 2003; Lee and Moray 1992; Lewandowsky et al. 2000)
Gong and Nass (2007)	–	–	<i>Objective</i> : behavioral trust (self-disclosure); <i>Subjective</i> : Individualized Trust Scale (Wheless and Grotz 1977)
Jensen et al. (2020)	“the willingness of a party to be vulnerable to the actions of another party based on the expectation that the other will perform a particular action important to the trustor, irrespective of the ability to monitor or control that party”	Mayer et al. (1995)	<i>Objective</i> : trust appropriateness (task allocation); <i>Subjective</i> : perceived trustworthiness (adapted from McKnight et al. (2020), originally from Mayer and Davis (1999))
Jensen et al. (2021)	“The willingness of a party to be vulnerable to the actions of another party based on the expectation that the other will perform a particular action important to the trustor, irrespective of the ability to monitor or control that party.”	Mayer et al. (1995)	<i>Objective</i> : trust appropriateness (task allocation); <i>Subjective</i> : perceived trustworthiness (adapted from Mayer and Davis (1999))
Jensen and Khan (2022)	“the willingness of a party to be vulnerable to the actions of another party based on the expectation that the other will perform a particular action important to the trustor, irrespective of the ability to monitor or control that other party.”	Mayer et al. (1995)	<i>Objective</i> : trust appropriateness (task allocation); <i>Subjective</i> : perceived trustworthiness (adapted from Mayer and Davis (1999))
Jung et al. (2019)	“the most dominant bases of trust are ability, persistence, and intention”	Lee and See (2004); Barber (1983); Dong et al. (2015); Mayer et al. (1995)	<i>Objective</i> : implicit trust (number of interventions, changes of trust in trial-level and judgement of trustworthiness (brain activity)); <i>Subjective</i> : ability, trust questionnaire (not specified)
Kahr et al. (2023)	“the attitude that an agent will help achieve an individual’s goal in a situation characterized by uncertainty and vulnerability” (Lee and See 2004)	Lee and See (2004)	<i>Objective</i> : behavioral trust (Weight on Advice); <i>Subjective</i> : single-item trust
Kulms and Kopp (2019)	“the attitude that an agent will help achieve an individual’s goals in a situation characterized by uncertainty and vulnerability” (Lee and See 2004)	Lee and See (2004)	<i>Objective</i> : behavioral trust (advise adoption); <i>Subjective</i> : single item (trust) and perceived trustworthiness (Fogg and Tseng 1999)
Muralidharan et al. (2014)	“the expectation of, or confidence in, another and is based on the probability that one party attaches to co-operative or favorable behavior by other parties”	Madhavan and Wiegmann (2007)	<i>Subjective</i> : trust scale (Jian et al. 1998)
Pak et al. (2012)	–	–	<i>Objective</i> : behavioral trust; <i>Subjective</i> : single-item trust on Likert-scale

Table 7 (Continued)
Tab. 7 (Fortsetzung)

Source (Author, Year)	Definition Trust	Reference	Measurement Trust
Park et al. (2023)	“an individual’s belief in a chatbot agent to help them achieve specific goals”	Lee and See (2004); Yagoda and Gillan (2012)	<i>Subjective</i> : trust scale (Lee and Choi 2017)
Smith et al. (2016)	–	–	<i>Objective</i> : behavioral trust (compliance)
Sobel and Sims (2022)	–	–	<i>Subjective</i> : perceived trustworthiness (Mayer and Davis 1999)
Tuncer and Ramirez (2022)	“the attitude that an agent will help achieve an individual’s goals in a situation characterized by uncertainty and vulnerability” (Lee and See 2004); “a sentiment resulting from knowledge, beliefs, emotions and other aspects of experience, generating positive or negative expectations concerning the reactions of a system and the interaction with it” (Hoffman et al. 2018)	Lee and See (2004); Hoffman et al. (2018)	<i>Subjective</i> : scenario-based trust scale (Ashoori and Weisz 2019)
Zou et al. (2020)	“an attitude towards an intelligent agent believing it can help achieve the user’s goal under uncertainty and vulnerability” (Lee and See 2004)	Lee and See (2004)	<i>Objective</i> : behavioral trust (compliance rate); <i>Subjective</i> : single item (trust)

Table 8 Key Findings of the Scoping-Review**Tab. 8** Zentrale Ergebnisse des Scoping-Reviews

Source (Author, Year)	Key Findings: Effect of Anthropomorphism on Trust
Abdulrahman and Richards (2022)	Anthropomorphism has no significant effect on trust (except for higher benevolence for human voice)
Cockram et al. (2023)	Perception of anthropomorphism not significantly different; Anthropomorphism has no significant effect on trust
de Visser et al. (2012)	Trust resilience increased trust calibration for anthropomorphic agents, over-reliance for machinelike agent
de Visser et al. (2016)	<i>Exp. 1:</i> trust resilience effect for anthropomorph agents, <i>Exp. 2:</i> lower initial trust, but higher trust resilience under greater uncertainty, <i>Exp. 3:</i> human repair behavior mitigates machinelike appearance, with no effect on trust calibration
de Visser et al. (2017)	Oxytocin augments trust only for reliable, anthropomorphic agents; lower trust resilience for anthropomorphic agents compared to human agents
Gong and Nass (2007)	Human stimulus is trusted more compared to a humanoid stimulus (matching face + voice); Consistency effect (consistent stimulus is trusted more than an inconsistent), more pronounced in females
Jensen et al. (2020)	More reliable systems are perceived as more humanlike; humanlike communication style is perceived as more benevolent (no effect for ability or integrity)
Jensen et al. (2021)	More reliable systems are perceived as more humanlike; Anthropomorphism does not influence trust appropriateness; higher perceived ability and integrity for anthropomorphic agents compared to computer agents
Jensen and Khan (2022)	Anthropomorphism does not impact behavioral trust; Anthropomorphism does not influence trust appropriateness; trust-dampening messages leads to higher perceived ability and benevolence only for computer agents
Jung et al. (2019)	Behavioral: Anthropomorphism has no significant effect on trust; but significant correlation between implicit trust and internal anthropomorphism (personality) on subject level; Brain activity: significant differences in trust evaluations for the external anthropomorphism (face)
Kahr et al. (2023)	Anthropomorphism (communication style) has no effect on trust (both subjective and objective); Anthropomorphism leads to higher behavioral trust only in high-accuracy models
Kulms and Kopp (2019)	Anthropomorphism has a (marginally) significant effect on trust (human agent is trusted more than the computer); higher perceived for trustworthiness for the avatar and human agent (with no difference between those)
Muralidharan et al. (2014)	Anthropomorphism has a significant effect on trust (perceived trust in the machine was significantly lower than for the human); trust ratings converge after task
Pak et al. (2012)	Anthropomorphism leads to higher (both subjective and objective) trust
Park et al. (2023)	Anthropomorphism leads to higher trust; human-like representations were perceived psychologically closer; higher intention to comply for human-like agent
Smith et al. (2016)	Anthropomorphism has no significant effect on trust in general; Higher compliance on the social task for the human agents; higher compliance for the mechanistic agents on the analytical task
Sobel and Sims (2022)	Anthropomorphism has no significant effect on trust
Tuncer and Ramirez (2022)	Anthropomorphism has no significant effect on trust
Zou et al. (2020)	Anthropomorphism has no significant effect on trust

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